Online Appendix for "Explanations"

Thomas Graeber   Christopher Roth   Constantin Schesch

June 3, 2026

This Online Appendix contains a full list of identified arguments (Section F) and additional results on confidence (Section G) and respondent characteristics (Section H) in our main experiment. Furthermore, it provides additional results on our richness manipulation experiment (Section I), the argument annotation experiment (Section J), binary Balls-and-urns (Section K) and continuous Balls-and-urns (Section L) experiments. Finally, it contains survey screens (Section M). The full text of all surveys is reproduced in an Online Supplement.

F List of identified arguments

Table F1 shows all naturally-occurring arguments in the 15 financial reasoning tasks we study in our main experiment, as identified by our annotation scheme detailed in Appendix B.

Table F1: Arguments Table

Argument	Description	Category
Task: Actively managed funds		
Active funds monitor & react to market	Actively managed funds can monitor and quickly adapt to market changes.	Fallacious
Impossible to predict stock market	Human inability to predict market movements, performance pressure, errors or overconfidence limit the effectiveness of active management.	Sound
Active funds managed by experts	Expertise in active management can lead to better investment decisions.	Fallacious
Active managers paid for performance	Active managers get paid because clients expect them to bring higher results than passive funds.	Fallacious
Active funds overperformed historically	References to historical data showing active management's performance.	Fallacious

Continued on next page

Argument	Description	Category
Passive funds overperformed historically	References to historical data showing passive management's performance.	Fallacious
Passive funds more stable, less risky	Passively managed funds maintain stability by not frequently changing investments, while actively managed funds are risky investments.	Sound
Passive funds more diversified	Passive management benefits from diversification across a broad market index.	Sound
Active funds charge fees	Investment fees of actively managed funds are higher than for passive management. They reduce net returns and negate potential gains.	Sound
Passive funds target long term	Passively managed funds tack market trends over the longer term, so that they are better at delivering long-term growth.	Fallacious
Passive funds track markets efficiently	Passively managed funds can achieve long-term growth by following market trends. Passive management is efficient in tracking market performance with minimal intervention.	Sound
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Bid ask spread		
Spread between bid & ask	Stocks have a bid and an ask price, and one can only buy the stock at the ask price which is always higher than the midpoint.	Sound
Buying stocks incurs fees	Buying stock through an online broker incurs additional fees, leading to a cost higher than the stock's listed price.	Sound
Quoted price is exact price	The cost of purchasing a stock is exactly the listed trading price if no fees are applied.	Fallacious
Taxes increase cost of stock	The cost of purchasing the stock is higher because of taxes.	Fallacious
Price has not changed since	Since the price hasn't changed since it was quoted, the stock can be bought at this exact price.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Crypto mining		
Resource intensity challenging for small miners	Bitcoin mining requires significant energy and resources, making it difficult for small miners. Large miners have an economic advantage in Bitcoin mining due to their scale and resources. Mining may not be profitable for small miners.	Sound
Mining by individuals still possible	Despite challenges, mining Bitcoin by individuals on a small scale is still possible, so that small miners dominate.	Fallacious
Decentralization different from equal distribution	Decentralization means that everyone can mine, but not that everyone mines equally, so that in practice larrge miners dominate.	Sound
Decentralization leads to small miners	Decentralization means there is no central planner, so that it leads to a diversity of miners, in which small miners dominate.	Fallacious
Shift from small to larger miners over time	There has been a historical shift from small miners to large mining operations over time.	Sound
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant

Continued on next page

Argument	Description	Category
Task: Disposition effect		
Sell depreciated stock for tax loss harvesting	Selling a stock that has lost value can be beneficial for tax purposes, allowing for tax loss harvesting.	Sound
Realizing loss means missing future gains	A stock that has lost value may have the potential to increase in value in the future, making it unwise to sell. Avoid selling stocks at a loss to prevent realizing the loss and potentially missing out on future gains.	Fallacious
Realizing gains of appreciated stock beneficial	Selling a stock that has gained value realizes the profit, ensuring a positive return on investment.	Fallacious
Stock will keep upward/downward momentum	One should keep the stock that has gone up and sell the stock that has gone down, because these trends can be expected to continue in the future.	Fallacious
Current gains or losses not predictive	Stock values fluctuate, so current losses or gains do not reflect future performance.	Sound
Higher value stock also more liquid	The stock with the highest value will also be more liquid, one should therefore sell that one.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Diversification		
Individual loss offset by other assets	Investing in multiple assets prevents total loss if one specific investment fails, akin to not putting all eggs in one basket.	Sound
Different assets respond differently to market	Different assets respond differently to market changes, so spreading investments can mitigate losses due to geopolitical or macroeconomic events.	Fallacious
Different assets respond similarly to market	Different assets usually respond similarly to market changes, so that it does not change much to invest in multiple assets instead of a single one.	Fallacious
Each asset is a chance to lose	Each asset is a chance to lose, so investing in multiple assets increases the chances of losing money.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Exponential growth bias		
Interest payments compound	The total amount in the savings account increases due to compound interest, where interest is earned on both the initial principal and the accumulated interest from previous periods.	Sound
Compute years times interest	A simple calculation of 2% interest per year on the initial 100, $leadingtoatotalof$ 110 after five years without considering compound interest.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Good company heuristic		
Higher growth brings higher returns	Investing in the firm with higher growth prospects will yield higher returns due to its potential for growth.	Fallacious

Continued on next page

Argument	Description	Category
Growth speculative & not guaranteed	Growth prospects are speculative and not a guaranteed indicator of future success, thus more information is needed.	Fallacious
More information needed	More information is needed to make a decision because the provided details are insufficient.	Sound
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Herding		
Future performance unpredictable	Past performance of cryptocurrencies does not guarantee future results. Timing the market correctly when investing in cryptocurrencies is not possible.	Sound
Own research necessary	It is important to conduct one's own research before investing in cryptocurrencies.	Fallacious
Risk of crypto requires caution	High volatility, risk of scams, lack of backing and other risks associated with cryptocurrencies are a reason for caution.	Fallacious
Friends may lack expertise	Friends providing advice may lack expertise in financial markets or cryptocurrencies.	Sound
Anecdotal evidence unreliable	Anecdotal evidence from friends is not a reliable basis for investment decisions, can be coincidence, luck etc.	Sound
Investments depend on individual circumstances	Investment decisions should be based on individual circumstances and not influenced by others. Cryptocurrencies may not be suitable for all investors.	Sound
Crypto potential for significant gains	Cryptocurrencies have the potential for significant gains from investing.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Historical stock returns		
Effect of inflation	Arguments that consider the impact of inflation on the average annual return.	Other
Relationship between volatility & returns	Arguments about how economic volatility affects the stock market's performance.	Other
Optimism about stock market	Arguments expressing a general optimism about the stock market's performance and long-term growth.	Other
Effect of general economic conditions	Arguments considering the general economic conditions and their impact on the stock market.	Other
Effect of specific historical events	Arguments considering the impact of specific historical economic events on the stock market, such as COVID-19 pandemic, recessions and subsequent recoveries etc.	Other
Anchoring on return during specific episode	Arguments where some remembrance of a specific or general stock returns is used as an anchor for the average return of the S&P 500.	Other
Known for high performance	The S&P500 is known for its high performance, which is why it has a historical average return above 10%.	Other
Known for being conservative	The S&P500 is known for being a popular, steady and conservative investment, which is why it has a historical return below 10%.	Other

Continued on next page

Argument	Description	Category
10% would be too high	Arguments based on the idea that a historical return above 10% seems too high. This can also involve the idea that, if that were true, everybody would be investing in the S&P500, which is not true and/or would reduce the return.	Other
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Home bias		
Company location irrelevant	The location of a company's headquarters does not impact its investment value.	Sound
Support local economy	Investing in a company headquartered in one's home state supports the local economy and community. This can also happen via taxes being paid in one's home state.	Fallacious
Local monitoring & access is easier	Investing in a company in one's home state allows for easier monitoring and access to the company.	Fallacious
Favorable tax implications	The choice between investing in a home state or out-of-state company may be influenced by different tax implications.	Fallacious
Preference for local company	A preference or bias towards investing in companies headquartered in one's home state.	Fallacious
Investments are identical other than location	Both investment options are considered equally good due to the companies being identical except for location.	Sound
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Interest rates and bond prices		
Inverse relationship between rate & price	Since there is an inverse relationship between interest rates and bond prices, bond prices will increase when the interest rate falls.	Sound
Increasing relationship between rate & price	Since there is a relationship in the same direction between interest rates and bond prices, bond prices will fall when the interest rate falls.	Fallacious
Fall in rates lowers demand	A fall in the interest rate leads to less demand and therefore a higher price of bonds.	Fallacious
Bond rates & prices unrelated	Bond prices remain stable and are not influenced by fluctuations in interest rates.	Fallacious
Lower rates mean lower coupons	Since the interest rate determines the interest payment that bondholders get from holding the bond, the bond's value will go down if the interest rate goes down.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Interest rates and stock prices		
Inverse relationship between rate & price	Since there is an inverse relationship between interest rates and stock prices, stock prices will increase when the interest rate falls.	Sound
Increasing relationship between rate & price	Since there is a relationship in the same direction between interest rates and stock prices, stock prices will fall when the interest rate falls.	Fallacious
Fall in rates lowers demand	A fall in the interest rate leads to less demand and therefore a lower price of stocks.	Fallacious
Higher company borrowing cost reduces stock price	Higher interest rates increase borrowing costs for companies, reducing their profitability and negatively affecting stock prices.	Sound

Continued on next page

Argument	Description	Category
Bonds & savings accounts become more attractive	Higher interest rates make bonds and savings accounts more attractive compared to stocks, leading investors to shift their investments.	Fallacious
Reduced consumer spending reduces profits	Higher interest rates reduce consumer spending (e.g. due to borrowing constraints), negatively affecting company profits and stock prices.	Sound
Raised cost of investments for investors	Interest rate increases raise the cost of investments, making it more expensive for investors and negatively affecting stock prices.	Fallacious
Rate hikes induce anxiety, reducing prices	Interest rate hikes make market participants uncertain and anxious, which reduces stock prices.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Nominal illusion		
Comparison between inflation & interest rate	Since the inflation rate is higher than the interest rate, one would be able to buy less tomorrow than today. This argument is distinct from PurchasingPowerDecrease because it displays no understanding of the mechanisms behind inflation and interest, and is based solely on a comparison of numbers.	Sound
Purchasing power decreases	Even though the amount of money in a savings account has increased thanks to the interest rate, the price at which one needs to buy goods and services will have increased more because of the comparatively higher inflation, so that the net effect on real purchasing power is higher. This argument is distinct from NumericalComparison because it displays an understanding of the mechanisms behind inflation and interest, not just a comparison of numbers.	Sound
Nominal spending higher thanks to interest	Because the amount of money in the savings account has increased thanks to the interest rate, one would be able to spend more than today.	Fallacious
Interest & inflation cancel each other out	Because the interest rate and inflation rate both cancel out, one would be able to buy exactly as much tomorrow as today.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Stock picking		
Everybody would do it	If it was possible to outperform the stock market by reading free online news, everybody would be doing it.	Sound
Markets are efficient	All publicly available information is already factored into stock prices, so that markets will already have adjusted to stale news.	Sound
News articles contain misinfo or bias	News articles can contain misinformation or bias, leading to poor investment decisions.	Sound
Market inherently unpredictable	The stock market is inherently volatile and unpredictable, making systematic outperformance difficult.	Sound
News insufficient, need expertise or intelligence	News are not enough for everyday people to outperform the stock market, since, for example, they also need to be specially smart, to have financial expertise and/or to have access to other sources of information.	Fallacious

Continued on next page

Argument	Description	Category
Any kind of effort or information pays	Any kind of effort, research or information will help to outperform the stock market.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant
Task: Value of call option		
Increases value because more upside potential	Higher volatility in a stock increases the potential for larger price movements, which can be advantageous for call option holders seeking to profit from upward stock movements.	Sound
Decreases value because more risk	Higher volatility is seen as increasing risk, making the call option less attractive and decreasing its value due to the unpredictability of stock price movements.	Fallacious
Option value determined by other factors	The volatility of a stock has no direct effect on the value of a call option because the call option's value is determined by other factors, not just the stock's volatility.	Fallacious
Other substantive argument	Any other substantive argument not part of the other categories.	Other
Irrelevant argument	Argument unrelated to the question; or no answer is implied by the argument.	Irrelevant

G Additional results on confidence

G.1 Differences in net learning rates

Figure G1 shows how imitation by receiver and orator confidence combine by showing the difference in net learning rate between *Explanation* and *Choice Only* by orator and receiver prior confidence. The net learning rate is defined as the difference in imitation rates between learning and unlearning situations. We split both variables by the aggregate median confidence of 73%. We find that confidence effects largely compound: the explanation effect on net learning is not significant when both orator and receiver are below the median; it is significant and of similar magnitude when either one is above the median; and it is highest when both are confident.

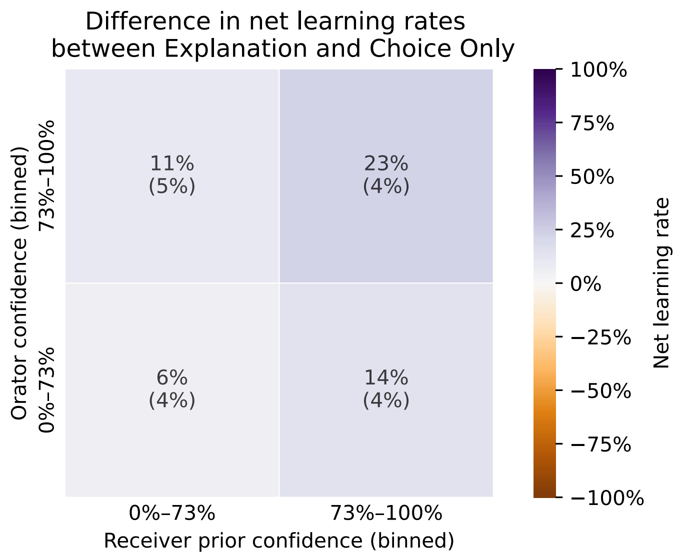
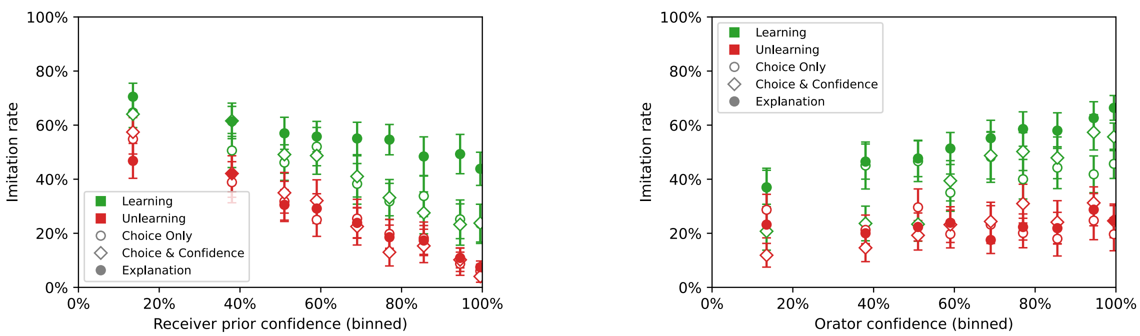


Figure G1: Effect of prior and orator confidence on imitation. *Notes:* Parentheses show SEs. *Explanation* sample is the main Receiver experiment (1,103 receivers, 13,111 obs.), *Choice Only* is pooled from all collections (2,642 receivers, 8,240 obs.).

G.2 Confidence and the content of explanations

We shed light on the mechanism behind these patterns by analyzing the interplay of confidence on the content of explanations.

We first repeat our previous analysis while also including the *Choice & Confidence* treatment. In Figure G2a, the effect of *Choice & Confidence* and *Choice Only* by receiver prior appears identical. Confidence signals do not help confident but wrong receivers adjust their answer. Although Figure G2b may appear to show a slightly larger effect of *Choice & Confidence* relative to *Choice Only* for confident orators in learning situations, the difference is not significant above ($p = 0.37$) or below ($p = 0.08$) median confidence. Numerical certainty statements therefore do not explain the effect of confidence, highlighting the importance of explanation content. [36]



(a) Imitation by prior confidence **(b)** Imitation by orator confidence

Figure G2: Effect of prior and orator confidence on imitation in *Choice & Confidence* treatment. *Notes:* See notes for Figure G1. Sample for *Choice & Confidence* is the corresponding receiver experiment (714 receivers, 8,531 obs.). Whiskers show 95% CIs with clustering at the orator and receiver levels.

We therefore turn to the role of orator confidence on content differences in explanations. Figure G3a plots the average number of high certainty markers (e.g., "I am certain that", "I am sure that") by orator confidence. The number of markers increases with confidence, but there is no clear difference between correct and incorrect explanations. Low certainty markers (e.g., "It might", "I'm not sure but"), shown in Figure G3b, conversely decrease with confidence, though right and wrong explanations do not significantly differ over most of the confidence range.

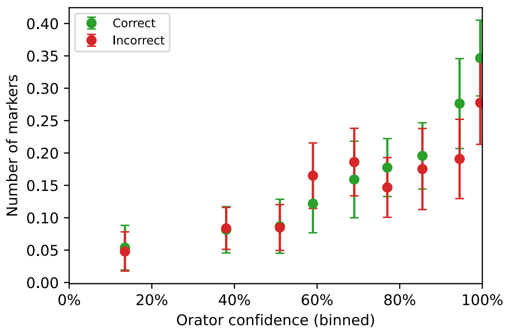
We then examine the effect of confidence on intellectual content by computing the shares of argument qualities in Figure G3c. The presence of sound arguments for correct explanations and of fallacious arguments for incorrect explanations clearly trend up in orator confidence. Other categories, and in particular no argument, in turn trend down, though there is no significant differentiating pattern between situations or confidence.

Finally, we focus on our main predictive variable by displaying richness by orator confidence in Figure G3d. We document three patterns. First, richness slopes up in orator confidence. Second, there is a richness gap between correct and incorrect explanations

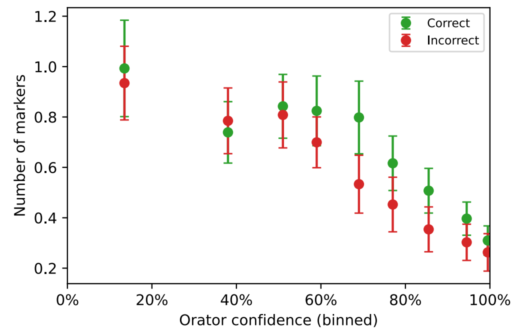
[36]Interestingly, even in the *Choice & Confidence* treatment which makes the latter very salient, there is no effect of orator confidence on imitation in unlearning situations.

at all levels of confidence. Third, this gap becomes larger when orators become more confident. More precisely, the richness gap between correct and incorrect explanations is 0.48 SD ($p < 0.01$) for orators with a confidence below the median against 0.91 SD ($p < 0.01$) with a confidence above it, a 0.42 SD difference ($p < 0.01$) which is slightly more than half the size of the unconditional richness gap.

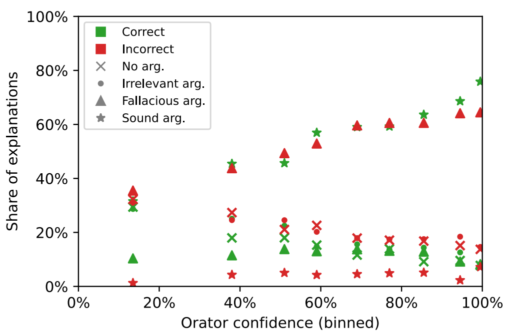
In conclusion, even within learning and unlearning situations, we uncover important differences in the content of explanations given by orators with differing certainties.



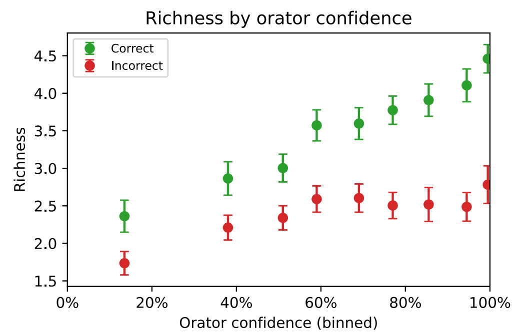
(a) High certainty markers by orator confidence



(b) Low certainty markers by orator confidence



(c) Argument quality by orator confidence



(d) Richness by orator confidence

Figure G3: Explanation content differences by orator confidence in learning and unlearning situations. *Notes:* Sample is the Orator experiment (466 orators, 6,910 obs.). For (c), see argument categorization in Section 5.2.1. Whiskers show 95% CIs with clustering at the orator level. In Figure (c), CIs are not shown for legibility, but all SEs are smaller than 4.4%.

G.3 Richness imitation and confidence

Just as there are substantial content differences driven by the orator side, receivers could differentially imitate explanations depending on their own and the orator's confidence. In particular, they could react differently to richness, which summarizes a broad range of explanation features.

Table G2 repeats our main decomposition, from Columns (1) and (7) in Table 1, separately for confident and unconfident receivers. Columns (1) and (3) reflect our earlier finding that explanations have a learning effect for both, but that it is about two times larger for confident receivers. Columns (2) and (4) show that the imitation of (standardized) richness is approximately the same for both, around 10 p.p., with an insignificant difference ($p = 0.10$). Since the richness gap between correct and incorrect explanations experienced by receivers is the same in both groups, richness therefore drives the same amount of learning asymmetry, of approximately $0.1 \times 0.744 = 7.4$ p.p.. This means it effectively explains the entire learning asymmetry for unconfident receivers, but only half of it for confident receivers. Put differently, confident receivers receive a benefit from explanations that is not explained by differential imitation of richness.

Turning to orators, we repeat the same decomposition separately for confident and unconfident orators in Table G3. Columns (1) and (2) again reflect our finding that explanations have a roughly two times larger learning effect when given by confident orators. Columns (2) and (4) establish that the imitation of richness is approximately the same in both samples in this case too, around 10 p.p., also with an insignificant difference ($p = 0.32$). Since we have found that the richness gap is 0.48 SD among unconfident orators but two times larger at 0.91 SD among confident ones, richness drives a two times larger learning effect among the latter. In turn, the residual not explained by richness is also roughly two times larger. In conclusion, the difference in explanation richness accounts for about half of the higher imitation of confident correct orators.

In conclusion, we have found that explanations have a learning benefit for confident and unconfident receivers, that this benefit is larger for more confident receivers, but that this effect is not driven by differential imitation of richness. It therefore reflects either content differences other than richness or a greater propensity to change their answer based on the arguments contained in explanations. In parallel, we have documented that explanations also have a larger learning benefit when they are delivered by confident orators, and that about half of this effect is driven by the fact that the richness gap between correct and incorrect orators is larger when they are more confident, which leads listeners to imitate correct and confident orators more.

Table G2: Richness imitation by prior confidence

	Dependent variable: Imitation			
	Receiver prior confidence ≤ 73%		Receiver prior confidence > 73%	
	(1)	(2)	(3)	(4)
Intercept	0.347***	0.339***	0.118***	0.122***
	(0.017)	(0.019)	(0.011)	(0.012)
Explanation	-0.008	0.055**	0.007	0.048***
	(0.022)	(0.023)	(0.014)	(0.016)
Learning	0.165***	0.175***	0.169***	0.163***
	(0.023)	(0.025)	(0.024)	(0.025)
Explanation × Learning	0.098***	0.013	0.196***	0.126***
	(0.030)	(0.031)	(0.030)	(0.031)
Richness		-0.016		0.008
		(0.013)		(0.012)
Explanation × Richness		0.122***		0.085***
		(0.016)		(0.017)
Observations	4849	4849	3940	3940
R^2	0.054	0.077	0.133	0.155

Notes: Sample is the main Receiver experiment for *Explanation* and all collections for *Choice Only*, both restricted to learning and unlearning situations. *Explanation* is a dummy for the *Explanation* treatment, *Learning* is a dummy for learning situations, *Richness* is standardized explanation richness. 73% is the median pooled orator and receiver prior confidence. We drop the 0.5% of observations with missing receiver prior confidence from all regressions. Standard errors are clustered at the orator and receiver levels.

Table G3: Richness imitation by orator confidence

	Dependent variable: Imitation			
	Orator confidence ≤ 73%		Orator confidence > 73%	
	(1)	(2)	(3)	(4)
Intercept	0.244***	0.226***	0.208***	0.210***
	(0.013)	(0.016)	(0.016)	(0.017)
Explanation	-0.029*	0.035*	0.036*	0.072***
	(0.017)	(0.020)	(0.022)	(0.022)
Learning	0.176***	0.192***	0.225***	0.221***
	(0.024)	(0.024)	(0.023)	(0.025)
Explanation × Learning	0.086***	0.030	0.156***	0.077**
	(0.029)	(0.030)	(0.030)	(0.032)
Richness		-0.033**		0.005
		(0.014)		(0.013)
Explanation × Richness		0.113***		0.089***
		(0.018)		(0.016)
Observations	4583	4583	4206	4206
R^2	0.061	0.073	0.124	0.144

Notes: See notes for Table G2.

H Additional results on orator & receiver characteristics

We examine how orator and receiver characteristics across learning and unlearning situations, as well as differential imitation based, e.g., on sociodemographics, may contribute to the asymmetric treatment effect of explanations.

We first document the heterogeneity of participants on observable characteristics. We then study which *orator* characteristics predict imitation, and whether this helps explain the asymmetric effect above and beyond the language differences associated with different groups of orators. We then test whether the characteristics of *receivers* predict their responsiveness to explanations, and to what extent this contributes to the asymmetric effect in the aggregate.

H.1 Participant characteristics in learning and unlearning

Because receivers and orators are drawn from the same population and because learning and unlearning are determined by the same endogenous variable—receiver prior accuracy—, orators in learning and receivers in unlearning situations *should* have the same characteristics on average; similarly, orators in unlearning and receivers in learning situations should be similar. Figure H4 shows the full set of observable characteristics we elicit in the study, for orators in the left panel and receivers in the right panel, separately for learning and unlearning situations.

Among orators, we find significant differences across three of the eight characteristics we examine between those in learning and unlearning situations. The first six characteristics capture participant sociodemographics. Orators with a correct answer have more education, are more likely to be male, less likely to be Black, and similarly likely to be Republican, to be older than 35 (the approximate median in our dataset) and to be working. The remaining two features characterize participants within the context of our study: orators with correct answers have a substantially higher prior accuracy rate in the 14 remaining tasks (61% vs. 52%) and a higher confidence on the present task (72% vs. 62%). Among receivers, we find very similar patterns for those with correct and incorrect priors, with significant differences in five out of eight features.

H.2 The role of orator characteristics

Heterogeneous treatment effects by orator characteristics. We ask whether orator characteristics drive imitation patterns. We begin by examining the raw relationship between orator characteristics and the treatment effect of explanations based on the *Choice Only* and *Explanation* conditions. In the left panel of Figure H5, we show the

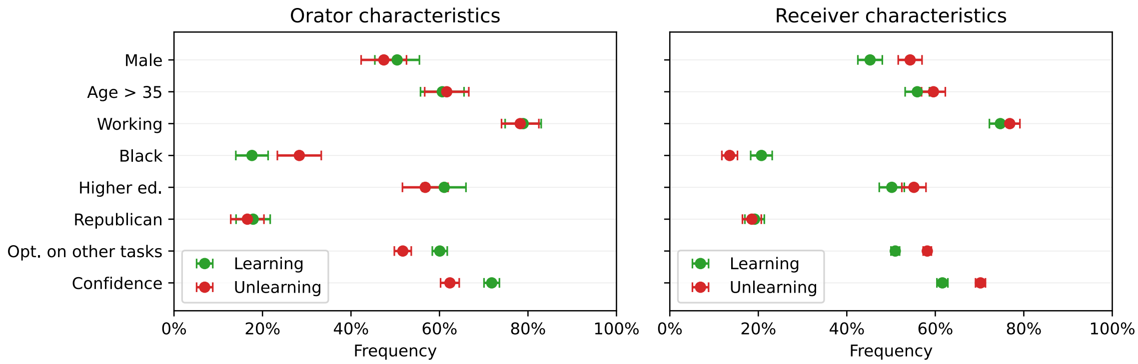


Figure H4: Characteristics of orators and receivers in learning and unlearning situations. *Notes: Optimality on other tasks* is the average optimality in the 14 other tasks. Confidence is rescaled from $[0, 100]$ to $[0, 1]$. 35 is the approximate median age in our data. Sample for both panels is the main Receiver experiment, matched with the Orator experiment for the left panel. Whiskers show 95% CIs with clustering at the orator and receiver levels.

effect of orator characteristics on imitation, controlling for the orator's and receiver's accuracy. We find that imitation is 2.2 p.p. lower for orators aged 35 or above ($p = 0.04$), 3.6 p.p. lower for African American orators ($p = 0.01$), and 2.4 p.p. higher for orators with higher education ($p = 0.02$). Point estimates further suggest that men are imitated more, but this is not significant. No receiver characteristics significantly predict imitation.

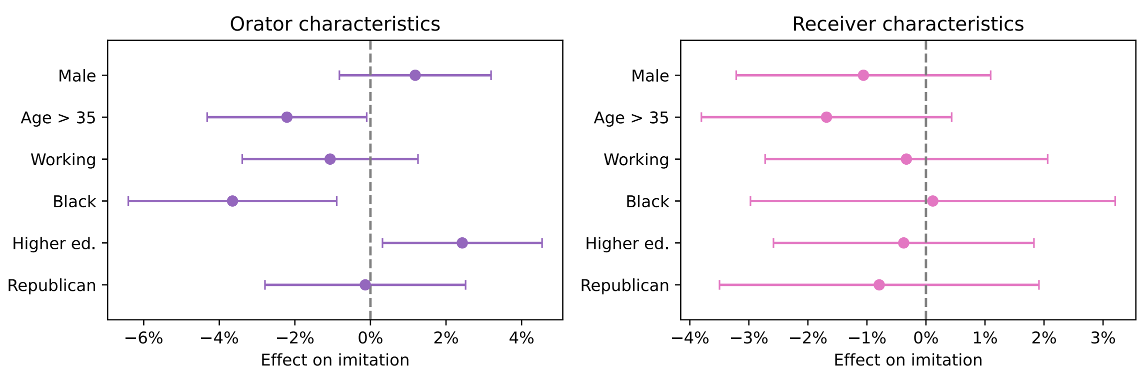


Figure H5: Effect of orator and receiver characteristics on imitation in *Explanation*. *Notes:* Left panel shows coefficients on *Explanation* interacted with orator characteristics, in a linear regression of imitation on orator and receiver optimality, *Explanation*, orator characteristics, and *Explanation* interacted with orator characteristics. Right panel shows the same results for receiver characteristics. Sample for *Explanation* is the main Receiver experiment, *Choice Only* is pooled from all collections. Whiskers show 95% CIs with clustering at the orator and receiver levels.

Do orator characteristics operate via content differences? Orator characteristics could drive imitation via their content, via their oral delivery, or because the orator's voice signals sociodemographic features that could influence imitation. To separate the role of content difference, we can focus on the transcript of each explanation and repeat the previous analysis while contrasting *Choice Only* with the *Transcript* treatment. In Figure H6, we can see that the effects of orator characteristics in *Transcript* are very close to those in *Explanation*, and that none of the differences between effects are statistically significant. This suggests that the effect of orator characteristics arises not from the orator's identity as revealed by speech, but from the different kinds of explanations different demographics use.

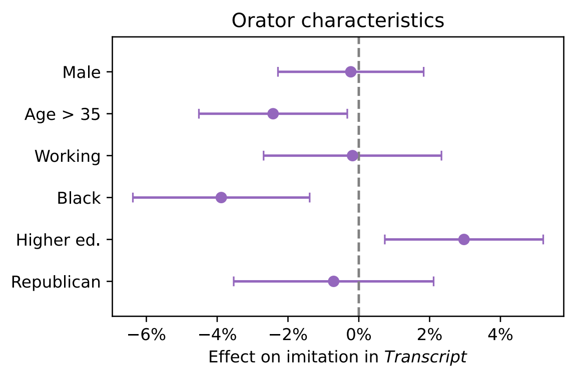


Figure H6: Effect of orator characteristics on imitation in *Transcript* treatment. *Notes:* See notes in Figure H5. *Transcript* sample is the corresponding Receiver experiment, *Choice Only* is pooled from all collections. Whiskers show 95% CIs with clustering at the orator and receiver levels.

Do orator characteristics explain away the asymmetric effect? Intuitively, orators in learning situations may signal characteristics through their voice, e.g., gender, which could make them more or less likely to be imitated. However, they also deliver different explanatory content. The preceding estimates should be interpreted as encapsulating *every* feature correlated with orator characteristics, both those contained in words and in their oral delivery.

Column 3 of Table 1 shows that orator characteristics explain away 32% of the differential effects, far below the 59% explained by richness. Moreover, columns 6 and 7 show that, once richness is controlled for, adding controls for orator characteristics does not further decrease the size of the unexplained learning effect. Figure H7 examines which orator characteristics explain away the differential effect. It shows that prior confidence and accuracy in the other tasks are most predictive, consistent with our finding that the effects of orator characteristics primarily operate through the content of explanations, though these effects are again very weak.

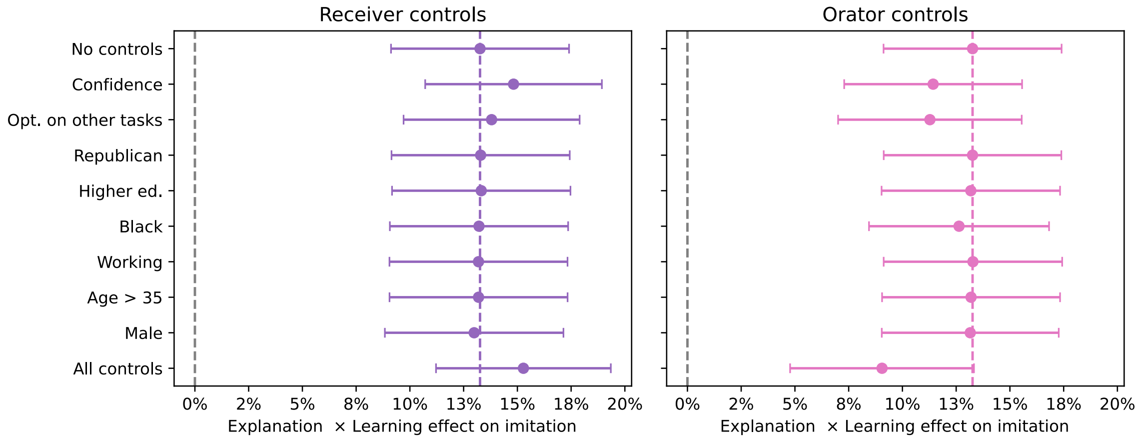


Figure H7: Learning asymmetry after controlling for orator or receiver characteristics. *Notes:* Coefficient on *Explanation* × *Learning* in a regression of imitation on *Explanation*, *Learning*, *Explanation* × *Learning*, *Control* and *Explanation* × *Control*. *Control* is a receiver control in the left panel and an orator control on the right. All regressions except 'No controls' and 'All controls' contain a single control. *Explanation* sample is the main Receiver experiment, *Choice Only* is pooled from all collections, both restricted to learning and unlearning situations. Whiskers show 95% CIs with clustering at the orator and receiver levels.

H.3 The role of receiver characteristics

Heterogeneous treatment effects by receiver characteristics. We plot the effect of receiver characteristics on imitation in the left panel of Figure H5. For example, after controlling for orator and receiver accuracy, a male receiver is 1.1 p.p. less likely to imitate. Again, it should be stressed that these effects are relatively weak since none of them are significant. Unlike in our analogous study of the orator side, the receiver analysis is not affected by content differences, since the content is controlled by orators, who are randomly assigned to receivers.

Do receiver characteristics explain away the asymmetric effect? From column 4 of Table 1, we can see that receiver characteristics, if anything, *increase* the asymmetric treatment effect between learning and unlearning. This is consistent with the idea that receivers with higher prior accuracy, i.e., those with unlearning opportunities, actually have a *better* assessment of whether another respondent's choice is right or wrong after listening to their explanation. Appendix Figure H7 studies the effect of controlling for each characteristic separately, finding that accounting for prior confidence and accuracy in the other tasks slightly increases this gap.

H.4 The role of interacted orator and receiver characteristics

We have studied orator and receiver characteristics separately, and found relatively mild effects, but it could be that their interaction produces strong effects. For example, men might be more likely to imitate other men, women might be less likely to imitate men, etc. This example shows that, for binary characteristics like ours, there are four relevant configurations, depending on whether the orator or receiver are in the in- or out-group.

For example, a male receiver is 2.5 p.p. less likely to imitate a female orator, while a woman is 1.9 p.p. more likely to imitate a man and a man is 0.1 p.p. more likely to imitate a man. All of these effects are relative to a woman listening to a woman, contrast *Explanation* with *Choice Only*, and control for orator and receiver accuracy. However, much like before, none of these effects appear strong enough to be significant.

Figure H9 repeats the same analysis but contrasts *Transcript* with *Choice Only*. Estimates are very similar and not statistically different from the previous one. This suggests that the effects of orator-receiver characteristics, if at all, seem to operate mainly through the content of explanations rather than pure signaling of socioeconomic characteristics.

In conclusion, although online speech-based surveys are promising to study bias in social learning or even discrimination, we do not find strong effects of orator or receiver characteristics when discussing financial reasoning tasks.

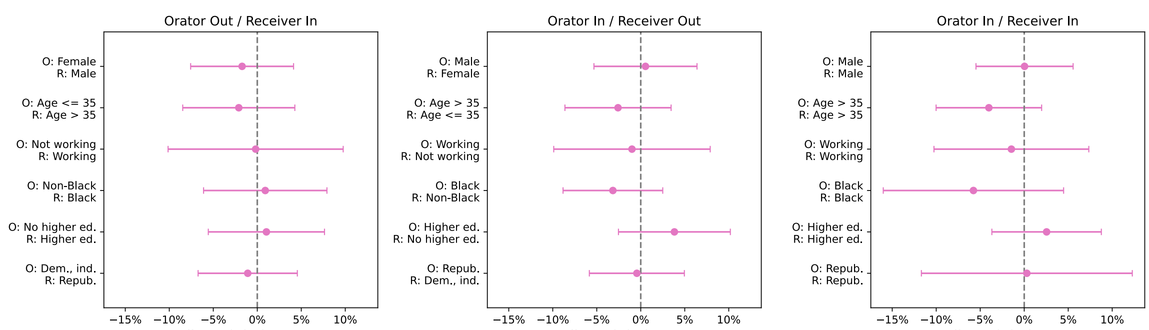


Figure H8: Effect of orator-receiver characteristics on imitation in *Explanation* treatment. *Notes:* Coefficients on *Explanation* interacted with orator-receiver characteristics, in a linear regression of imitation on orator and receiver optimality, *Explanation*, orator-receiver characteristics and *Explanation* interacted with orator-receiver characteristics. *Explanation* sample is the main Receiver experiment, *Choice Only* is pooled from all collections. Orator-receiver characteristics are an *Orator Out / Receiver In* dummy equal to 1 if the receiver has the characteristic but not the Orator; *Orator In / Receiver Out* and *Orator In / Receiver In* are analogously defined; *Orator Out / Receiver Out* is left out and serves as reference level. Whiskers show 95% CIs with clustering at the orator and receiver levels.

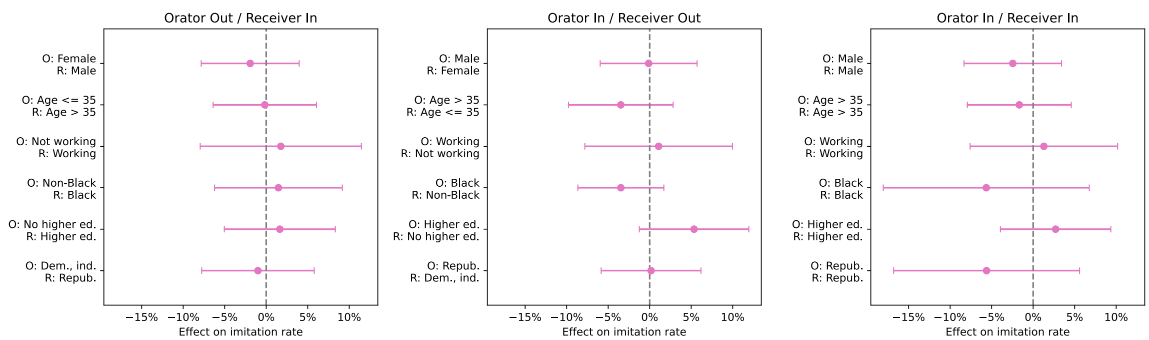


Figure H9: Effect of orator-receiver characteristics on imitation in *Transcript* treatment. *Notes:* See notes in Figure H8. *Transcript* sample is the corresponding Receiver experiment, *Choice Only* is pooled from all collections. Whiskers show 95% CIs with clustering at the orator and receiver levels.

I Richness manipulation experiment

This Section summarizes additional details about the experiment exogenously manipulating linguistic richness.

I.1 Details on the generation of rich & sparse versions

Our experimental design aims to take each explanation given by a respondent and generate a rich and a sparse variant that (i) preserve the informational content, and (ii) remain as close as possible to the natural tone of respondents.

We address (i) by heavily emphasizing in our prompt that the informational content, and in particular "all arguments, facts or references" should be exactly preserved. An inspection of original explanations and generated versions shows this works quite well. The ex-post argument identification exercise described below confirms this finding.

Achieving (ii) is more challenging because LLMs are pre-trained on corpora of written text and fine-tuned to be maximally intelligent and helpful assistants. We therefore give the model examples, for each question, of 20 explanations randomly drawn from the 50 with the lowest richness rating (to avoid having only extremely sparse explanations), and the 5 with the highest richness rating (since these are longer).

We then prompt it to generate a maximally sparse and a maximally rich version by imitating the style of the examples while keeping the informational content constant. We instruct the LLM to pay close attention to specific features when imitating the sparse and rich examples. These features are informed by our analysis of the content of explanations and of the determinants of richness (cf. Section 5.2).

To generate the versions, we used GPT-4.1 (more specifically, `gpt-4.1-2025-04-14`), OpenAI's "flagship model for complex tasks" as of June 2025. Below, we reproduce the full prompt used to generate both versions at once:

> Survey respondents were asked the following financial literacy question: {question}
>
> They also recorded oral responses to explain their choice. A rich explanation is detailed, comprehensive, logically structured, nuanced, and tailored to the context. A sparse explanation is basic, narrow, unclear or disorganized, presents only surface-level understanding, lacks depth or specific details, and fails to relate clearly to the context. Richness is measured as an integer between 0 and 10, both inclusive. If the explanation provides no argument, it is 0. Richness should not depend on whether the respondent's answer or their premises are factually correct—only the quality of explanation matters.
>
> To give some examples, here are twenty maximally sparse and five maximally rich explana-

tions given by respondents:

{20 sparse examples}

{5 rich examples}

We now focus on a specific respondent, who chose the following answer: {answer}

The respondent also gave the following oral response to explain their choice: {speech}

Please produce two versions of the above explanation, one that is maximally sparse (richness score: 1/10) and one that is maximally rich (richness score: 9/10). Each version should exactly preserve the informational content of the original explanation. In particular, it should preserve all arguments, facts or references of the original explanation, and not add to them. Instead, your re-writings should focus on structure, coherence, presentation and wording of the explanation, and very closely imitate the corresponding examples given above.

Both versions should closely imitate the oral style of corresponding explanations. In particular, you should pay close attention to filler words ("uh", "um", "huh"), false starts ("So, uh, I think, maybe..."), repetitions and restarts ("I believe that, um, I believe that..."), rhetorical questions ("Why do they outperform?") and first-person framing ("I believe...") and meta-commentary ("This is an easy/difficult question", "To answer this question", "To maximize your chances of getting it right" etc.).

The maximally sparse version should not be a pure summarization of the explanation. When imitating the given examples, you should pay close attention to contradictions or self-corrections mid-sentence ("I think it's one—no wait, maybe two..."), vague or circular language, replacement of specific vocabulary with informal terms ("stuff", "thing"), fragmented or run-on sentences ("So I say one, because, like, I think..."), clauses connected without logical links ("I think, it has to be that, I mean clearly"), abrupt cutoffs or trailing thoughts ("So yeah, number one, because they..."), disconnected or illogical flow, introduction of arguments without relating to the overall context and over-reliance on the prompt.

The maximally rich version should not be a mere rewording of the explanation. When imitating the given examples, you should pay close attention to their clear logical sequence and explicit connectives ("because", "therefore", "for instance"), clarification of underlying assumptions and concepts, specific vocabulary and avoidance of vague words, illustrative examples and real-world analogies, contextual framing of arguments ("Let's say the interest rate is 2%..."), and structure designed to help someone else choose the correct answer rather than just restating their own.

I.2 Richness of rich & sparse versions

We display the distribution of richness scores in Figure I10, which shows that the sparse versions are indeed sparser and the rich versions richer than the originals. Moreover, all distributions appear relatively well-behaved, with no extreme clustering at either end.[37]

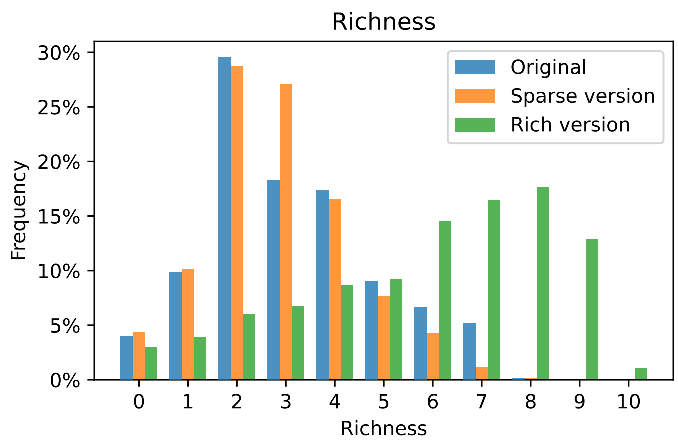


Figure I10: Distribution of richness of original, rich & sparse versions. *Notes:* Sample is the Orator experiment (466 orators, 6,910 obs.) and variants generated with an LLM.

We can characterize the effect of the richness manipulation depending on the accuracy of the orator's answer. When the orator is correct, original richness is 3.74 on average, against 3.38 for *Sparse version* ($p < 0.01$) and 6.74 for *Rich version* ($p < 0.01$). On the other hand, when the orator is incorrect, average richness is 2.40, 2.24 ($p < 0.01$) and 4.73 ($p < 0.01$), respectively.

Finally, we examine how each version's richness depends on the original's richness in Table I4. Column 1 shows that the sparse version dampens original richness but has a slight fixed boost, while column 3 shows the rich version amplifies original richness differences and also has a much more sizeable fixed richness boost. Columns 2 and 4 show that there is no strong effect of orator accuracy on this pattern for sparse versions, but that rich versions amplify richness heterogeneity less strongly for correct orators. Perhaps this is because correct explanations are often already comparatively rich, so that there is less room to increase it further.

[37]This well-behavedness is somewhat at odds with our prompt, which instructed the LLM to aim for a richness of 1/10 and 9/10, respectively. Experimentation shows LLMs are not very good at obtaining very specific richness scores, so that these indications were only meant to broadly increase richness differences.

Table I4: Effect of original richness on sparse and rich version's richness

	Sparse version richness		Rich version richness	
	(1)	(2)	(3)	(4)
Intercept	0.657***	0.649***	2.212***	1.487***
	(0.024)	(0.033)	(0.067)	(0.070)
Orator correct		0.157***		1.571***
		(0.040)		(0.093)
Original richness	0.705***	0.662***	1.157***	1.353***
	(0.008)	(0.013)	(0.018)	(0.026)
Original richness × Orator correct		0.026*		-0.370***
		(0.014)		(0.025)
Observations	6910	6910	6910	6910
R^2	0.701	0.706	0.650	0.671

Notes: All richness measures are standardized using the mean and standard deviation of original explanations. Sample is the Orator experiment (466 orators, 6,910 obs.) and variants generated with an LLM. Standard errors are clustered at the orator and receiver levels.

I.3 Features of rich & sparse versions

To shed more light on what is driving this exogenous linguistic richness variation, we analyze the features of the generated versions in more depth. We therefore repeat the feature annotation exercise described in Section 5.2 separately for each version, and display results in Figure I11.

Beginning with certainty markers, we find that both rich and sparse versions contain more low-confidence markers than original explanations, meaning that even in rich versions the LLM communicates more extensively about the uncertainty contained in the initial explanation. Sparse explanations contain fewer high confidence markers and even rich versions contain slightly fewer than the originals, perhaps because the LLM is slightly overcautious in wanting to preserve informational content.

Disfluencies are more common in sparse versions than in the original and rarer in rich versions, a pattern that applies to self-corrections, repetitions, false starts and especially strongly to filled pauses. Indeed, originals contain 1.06 filled pauses (e.g., "uh" or "uhm") against only 0.17 in rich versions and 2.89 in sparse versions. Clearly, the LLM identifies these as prominent features of maximally rich or sparse explanations. In most other features, we also obtain that they are less frequent in sparse versions and more frequent in rich ones.

Taken together, these results show that the linguistic richness manipulation affects the presence of numerical correlates of richness and of language markers that contribute to it. We therefore conclude that it achieved its aim in an effective, broad and balanced way.

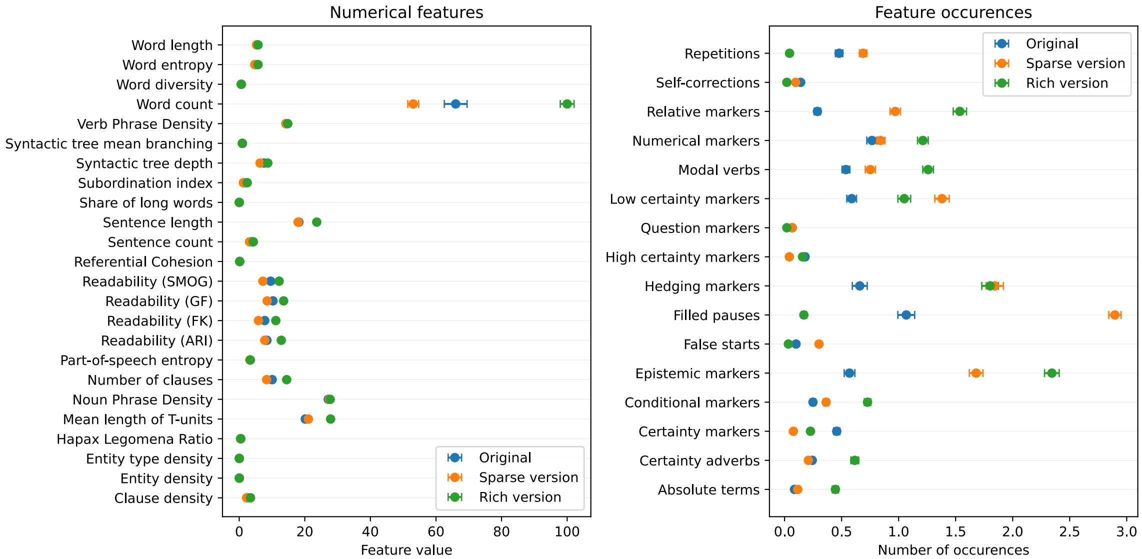


Figure I11: Features and markers of original, rich & sparse versions. *Notes:* Sample is the Orator experiment (466 orators, 6,910 obs.) and variants generated with an LLM.

I.4 Arguments of rich & sparse versions

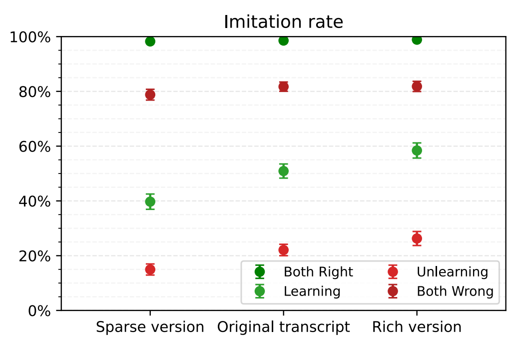
To verify that each version indeed preserves the informational content of explanations, as is stressed in the LLM prompt, we repeat the argument-tagging exercise described in Section 5.1.1 independently for each version. More precisely, we use GPT-4 again to tag which of our manually identified arguments are present in each edited version, separately for each version of each explanation.

Across all rounds and all arguments (including "Other" and "Irrelevant"), the correlation between arguments identified in the original explanation and in the sparse version is 0.68, with a Cohen's κ of 0.67 indicating "substantial agreement". Similarly, the correlation between arguments in the original and the rich version is 0.66, and a Cohen's κ of 0.66 again reflecting "substantial agreement". Agreement between the two versions, the condition which ensures the validity of our experimental design, is even higher with a correlation of 0.76, and a Cohen's κ at 0.76 showing we have "substantial agreement" and are very close to "almost perfect agreement".

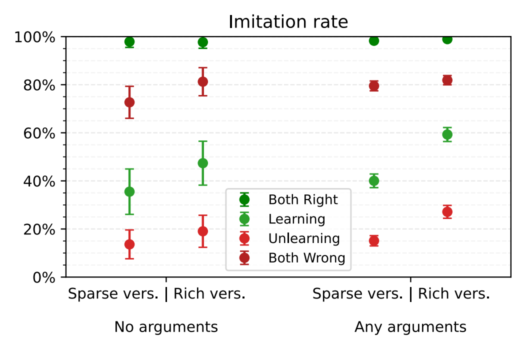
When an argument is not identified in the rich version, it is also not identified in the sparse version in 98% of cases. However, when an argument is identified in the rich version, it is also identified in the sparse version 73% of the time. Indeed, sparse versions contain 1.09 arguments on average, slightly less than the rich versions which contain 1.32, a comparatively small but significant difference ($p < 0.01$).

In conclusion, the richness manipulation appears very effective at preserving the informational content of explanations, though it is not perfect. It should be noted that assessing the presence of specific arguments is inherently noisy, so that even preserving informational content perfectly is likely to lead to some disagreement in the ex-post argument identification. Moreover, since rich versions contain more words and LLMs work on tokenized words, there is a higher chance that the LLM picks up on an instance of an argument for richer transcripts, so that the small differences we do observe could reflect this algorithmic bias.

Nonetheless, as a conservative robustness check, anticipating on Section 5.3.3, we repeat our main analysis on the explanations for which there is complete agreement on the informational content of both versions. For 63% of the sample, GPT-4 exactly agrees on the presence or absence of each of the arguments, including "Other" and "Irrelevant", in both versions. This fraction is quite high given the noisy nature of argument tagging, with another substantial fraction of explanations agreeing on all but one or two arguments. Among this sample with perfect agreement, imitation is 7.1 p.p. higher in *Rich version* than in *Sparse version* ($p < 0.01$). Moreover, it is 18.6 p.p. higher in learning situations ($p < 0.01$), 8.6 p.p. higher in unlearning situations ($p < 0.01$), 2.9 p.p. higher when both are wrong ($p = 0.05$) and 0.6 p.p. higher when both are right ($p = 0.22$). Since these effects are significant and very close to those in the full sample, we conclude that our experimental findings are not driven by changes in the informational content of explanations.



(a) Imitation of original, rich & sparse versions



(b) Imitation by argument presence

Figure I12: Effect of richness manipulation on imitation, relative to *Transcript* and by presence of arguments. *Notes:* Sample is the Richness receiver survey (972 receivers, 14,580 obs.). Whiskers show 95% CIs with clustering at the orator and receiver levels.

I.5 Heterogeneity by presence of arguments

Another check we can implement is to test whether the richness manipulation has effects that differ based on the presence of any argument. Large differences in the effect of the treatment between explanations with and without arguments suggest that variation in informational content plays some role in the results. For explanations without any arguments, imitation is 57.0% and 62.1% in *Sparse version* and *Rich version*, respectively, against 65.0% and 72.8% for explanations with arguments. This 2.7 p.p. difference in effect sizes is not very large and not statistically significant ($p = 0.37$). Figure I12b plots the imitation rates for each condition separately for each configuration, which also all have small and insignificant differences in effect sizes. However, it should be noted that this test is not very powerful because explanations without any arguments are quite rare, making up only 7.9% of the sample.

I.6 Comparison with original *Transcript* treatment

Although results should be interpreted more carefully since they are not part of the same experimental design, it is nonetheless interesting to compare imitation of the edited versions with that of the original explanations in the *Transcript* condition. Overall, it sits between the sparse and rich versions, since its imitation rate is 69.5%, above the 64.3% of *Sparse version* ($p < 0.01$) and below the 71.9% of *Rich version* ($p < 0.01$). Figure I12a documents that, also within each situation, original explanations are in-between the sparse and rich versions. Since the richness of originals also sits in-between that of the versions, these findings further point toward an effect of richness on imitation.

J Argument annotation experiment

J.1 Design & logistics

In the instructions, participants were told that previous respondents had answered financial reasoning questions and recorded oral explanations intended for future participants. These explanations had been transcribed and edited. The task was to read 10 such explanations for a single task and, for each one, select all arguments from a provided list that were present in the explanation. Participants were explicitly instructed that an explanation could contain multiple arguments, one argument, or none of the listed arguments.

The instructions emphasized that arguments should be identified based on their logical substance rather than their wording. Participants were told to select an argument when the explanation expressed the same reasoning, even if it used different words, and not to select an argument merely because it contained similar vocabulary. The list of arguments in each task is as in Table F1, with the description being presented to participants. Figure M26 shows the annotation screen.

Participants received a base reward of $5. To incentivize careful coding, receivers had a 10% chance of being eligible for a $10 bonus with "beauty contest" incentives. If the receiver was eligible for the bonus, one of their annotated explanations and one of its arguments were randomly selected, and the participant entered a lottery with winning probability equal to the share of other participants whose classification matched theirs.

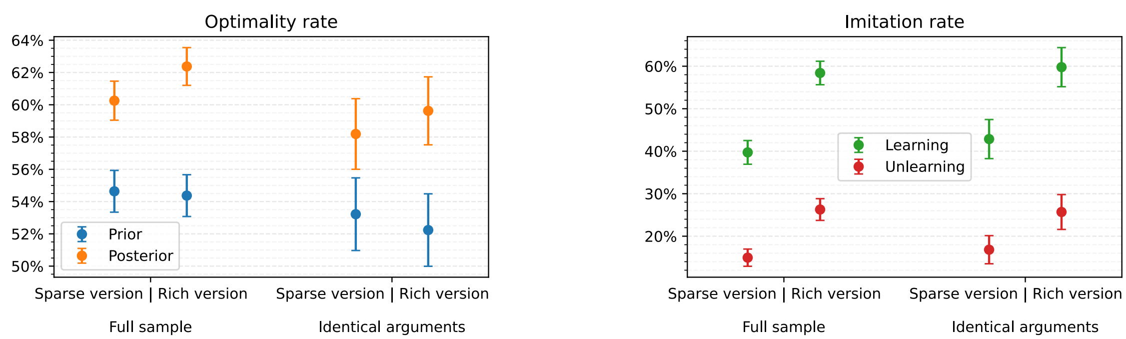
The experiment was run in May 2026 using a U.S. sample on Prolific. Median completion time was 16 minutes. Out of 1,990 respondents, 1,967 gave valid data. The quota system distributed explanations across annotators, while ensuring that no respondent saw both the rich and the sparse version of the same explanation. In line with our preregistration, we discard the first two out of ten annotations so annotators could get used to the arguments. We then select exactly one annotation per speech-version, and select speeches for which each version has an annotation. The final sample hence contains one annotation for the rich and sparse versions of 6,883 explanations, representing 99.6% of our total sample of 6,910.[38]

[38]The discarded annotations, coming notably from the first two rounds, yield multiple annotations per version-pair, which were used to implement the "beauty contest" incentives.

J.2 Richness effect in subsample with identical arguments

Across all 6,883 explanations, annotators identified exactly the same set of arguments in the rich and sparse version in 2,069. This represents an exact agreement share of 30.1% (95% CI: 28.9%–31.3%).

We verify whether the treatment effect of the richness manipulation experiment holds in this subset. Figure J13 plots the effects of the richness manipulation on imitation and optimality, for the full sample (as in Figure 9) and the agreement subsample. The effect on optimality, already muted in the full sample, loses significance ($p = 0.26$) with wide confidence intervals in the subsample. However, the effect on imitation remains strongly significant at +8.9 p.p. ($p < 0.01$) in unlearning and +16.9 p.p. ($p < 0.01$) in learning situations. We also find no evidence that the effect is significantly different between the two samples for optimality ($p = 0.56$) or imitation ($p = 0.23$). The exogenous richness evidence is therefore robust to keeping the informational content of explanations exactly constant between rich and sparse versions.



(a) Optimality by version and agreement

(b) Imitation in learning and unlearning

Figure J13: Optimality and imitation in the identical-argument subsample. *Notes:* See notes for Figure 9. *Identical arguments* denotes the subsample of explanations where respondents identified the same set of arguments in the rich and sparse version. Whiskers show 95% CIs with clustering at the orator and receiver levels.

J.3 Comparison of human and LLM annotation

We investigate agreement between the human annotation in this experiment and the annotation performed with GPT-4 in Section I.4. Inter-rater reliability is high. If GPT-4 identifies a specific argument, there is a 66% chance the human annotator does so as well. If GPT-4 does not identify an argument, there is a 92% chance the human does not either. Cohen's κ between the two is 0.59, indicating "moderate agreement" and very close to "substantial agreement". Inter-rater reliability is again even higher for argument categories, with positive and negative agreement rates of 88% and 46% for "any argument", 83% and 88% for "any sound argument", 77% and 83% for "any fallacious argument", 18% and 98% for "any irrelevant argument". The latter is a notable exception, and reflects the fact that humans classify 13.5% of explanations as containing an irrelevant argument, while GPT-4 is much more generous at only 3.9%. We view these benchmarks as confirming the overall quality of the respondents' annotation. We find no evidence of a difference in treatment effects between the human and GPT-4 identical-argument subsamples ($p = 0.254$).

Finally, we investigate how richness affects the argument annotations of humans and GPT-4. Table J5 regresses the number of arguments identified by humans and GPT-4 on the version. Column (1) shows human annotators from Prolific identify 0.997 arguments on average in sparse versions, and 0.275 arguments more in rich versions ($p < 0.01$). Conversely, column (2) shows GPT-4 identifies 0.955 arguments in sparse versions and 0.188 ($p < 0.01$) arguments more in rich versions. Therefore, as is shown in column (3) where the outcome is the difference in the number of arguments identified by respondents and GPT-4, humans identify 0.042 arguments more in sparse versions ($p < 0.01$), and an additional 0.087 more in rich versions ($p < 0.01$). Humans are therefore more strongly influenced by richness than GPT-4 when identifying arguments.

Table J5: Number of arguments identified by humans and GPT-4 in rich and sparse versions

	# arguments (humans) (1)	# arguments (GPT-4) (2)	Δ # arguments (3)
Intercept	0.997***	0.955***	0.042***
	(0.018)	(0.017)	(0.014)
Rich version	0.275***	0.188***	0.087***
	(0.013)	(0.007)	(0.014)
Observations	13766	13766	13766
R^2	0.021	0.013	0.003

Notes: Δ is the difference in the number of arguments annotated by humans and GPT-4. Sample is a subset of the richness manipulation experiment (1,963 receivers, 13,766 obs.). Standard errors are clustered at the orator (i.e., explanation) and annotator levels.

J.4 Channels of richness imitations

Richness could increase imitation through two channels. First, it could increase the perceived quality and number of arguments in an explanation. Second, it could lead receivers to imitate more even holding the logical content fixed, perhaps because linguistic richness acts as signal about the speaker's competence.

We can cautiously approach this question by investigating how the effect of the linguistic richness manipulation evolves as we add finer and finer controls for explanations' logical content. Column (1) of Table XYZ regresses imitation in learning and unlearning situations on the rich version, showing our main 15.4 p.p. treatment effect ($p < 0.01$). Column (2) adds a control for the rich or sparse versions having different perceived argument sets in the annotation experiment, recovering the 13.4 p.p. effect in the agreement subset ($p < 0.01$), which represents 87% of the total effect.[39] Column (3) controls for the number of arguments identified in each version in the annotator survey, showing that the rich version treatment still induces a 14.2 p.p. effect ($p < 0.01$), i.e. 92% of the total. Finally, column (4) adds a fixed effect for each of the identified arguments, after which the richness manipulation nonetheless increases imitation by 13.7 p.p. ($p < 0.01$), making up 89% of the total effect.

These findings are compatible with the view that richness affects both the perceived informational content of explanations and imitation conditional on informational content, and that a large share of its total effect is probably driven by the latter. However, it should be stressed that this is only very imperfect evidence, as it is very difficult to proxy the informational content of explanations as perceived by each listener. Moreover, here, variation in perceived argumentative content are driven by residual differences across versions written by the LLM, which is presumably very different from variations in natural settings.

In conclusion, the annotation experiment documents that richness robustly affects imitation both via the perceived informational content of explanations and via linguistic richness conditional on the informational content, both of which drive social learning.

[39]Since *Different arguments* varies at the level of a rich–sparse pair, and we pool across pairs, the coefficient on *Different arguments* captures the imitation effect of having a different argument set for sparse versions, which is (insignificantly) negative, while the coefficient on *Rich version* × *Different arguments* captures the effect on rich explanations, which is (insignificantly) positive. This reflects the fact that rich versions are judged to have slightly more arguments and are hence imitated more.

Table J6: Effect of richness manipulation conditional on argument content of explanations

	Dependent variable: Imitation			
	(1)	(2)	(3)	(4)
Intercept	0.279***	0.292***	0.224***	0.268***
	(0.011)	(0.016)	(0.012)	(0.017)
Rich version	0.154***	0.134***	0.142***	0.137***
	(0.013)	(0.021)	(0.013)	(0.013)
Different arguments		-0.020		
		(0.017)		
Rich version *imes* Different arguments		0.030		
		(0.025)		
# arguments			0.056***	
			(0.007)	
Argument FEs				✓
Observations	6036	6036	6036	6036
R^2	0.026	0.026	0.037	0.131

Notes: Rich version is a dummy for the rich version treatment. *Different arguments* is a dummy for whether the rich and sparse versions Δ is the difference in the number of arguments annotated by humans and GPT-4. Sample is a subset of the richness manipulation experiment (1,963 receivers, 13,766 obs.) restricted to learning and unlearning situations. Standard errors are clustered at the orator and receiver levels.

K Binary Balls-and-urns experiment

K.1 Design and logistics

In the Orator experiment, participants record an explanation, select their answer and state their confidence. In the Receiver experiment, a separate sample of participants is given instructions of the task with an example. They first state their prior responses and their confidence. They then either (i) learn about another respondent's choice (*Choice Only*) or (ii) additionally listen to a verbal explanation behind the choice (*Explanation*). Subsequently, respondents again state their choice and confidence. Receivers have a 40% probability of being assigned to the *Choice Only* treatment and a 60% probability of being assigned to the *Explanation* treatment.

Respondents in both studies received a base reward of $2 for completing the study. Participants could win a bonus of $10 following the same incentive structure as in the main experiment. Median completion times were 11 and 10 minutes in the Orator and Receiver experiment respectively. Both experiments were run in May 2026 using a U.S. sample on Prolific. The Orator experiment was run for a total of 552 respondents, of whom 483 gave valid data. The Receiver experiment was run for 2,006 respondents, of whom 1,914 gave valid data.

In line with our pre-registration, we drop respondents who self-reported searching for answers online or using external tools. To curb the use of AI assistants,[40] we asked receivers whether the orator used an automatic voice generator or seemed to be reading a script. As specified in our pre-registration, we then exclude all orators for whom a majority of receivers agree: these cases represent 4.4% of valid recordings.

Due to the binary elicitation, we expected the optimality rate to be high. This implies learning and unlearning situations should be very rare in the sample under random matching. To increase power, we therefore stratify the orator assignment based on the receiver's prior answer. If a receiver's prior answer is correct, we randomly assign them to a correct or incorrect orator with equal probability. If it is incorrect, we assign them an orator without stratification. In our raw sample, receiver and orator are both right in 63.2% and both wrong in 9.0% of observations, while learning and unlearning situations occur 9.5% and 18.3% of the time. We then de-stratify our sample using expected

[40]In all experiments, the question text is not selectable and therefore difficult to copy-and-paste into a search engine or AI assistant. However, between the main experiment in December 2023 and additional experiment in April 2025 and May 2026, multimodal AI assistants that can analyze screenshots have become much more prevalent. We believe this explains the noticeable rise in orators that appear to read a script from an AI assistant.

frequencies under random matching, with an optimality rate for each configuration estimated by pooling orators and receiver priors. All summary statistics, regressions and figures presented in the main body and in the following sections are re-weighted to the population frequency.

K.2 Optimality and imitation

Answers are correct if they say which bag is more likely in line with the Bayesian posterior. Figure 11 displays optimality rates and imitation rates for the balls-and-urns task. Prior optimality stands at 83.6% and 80.3% in *Choice Only* and *Explanation*, respectively ($p = 0.03$), which, as expected, is relatively high. This reflects the binary elicitation scheme, which considerably simplifies the task. In particular, when the ball draw signal goes in the same direction as the prior, the task becomes essentially trivial and optimality rates very close to 100%. As in our main experiment, both treatments help listeners select correct answers and significantly increase their optimality rates by 1.9 p.p. ($p = 0.01$) and 3.7 p.p. ($p < 0.01$) respectively. *Explanation* hence increases the posterior optimality rate by 1.8 p.p. more ($p = 0.09$) than *Choice Only*.

Figure 11 also plots imitation in learning and unlearning situations. The share of receivers imitating the orator is 5.5% and 5.2% for *Choice Only* and *Explanation* in unlearning opportunities ($p = 0.93$). In learning opportunities, it jumps from 27.0% to 51.6%, a highly significant treatment effect ($p < 0.01$). As in our main experiments, explanations therefore increase imitation in learning but not unlearning situations.

K.3 Imitation of arguments and richness

We now analyze the substantive content of explanations in this experiment. Our binary elicitation implies that, while under *Choice Only* receivers only see which ball was deemed more likely by the orator, under *Explanation* the orator could also state a numerical posterior belief. As pre-registered, we use GPT-4 to identify which explanations state a numerical posterior. We specifically instruct it to only retain posterior belief statements, not pure repetition of numbers from the question. 10.9% of explanations explicitly state a numerical posterior belief, a relatively low share. Moreover, 43.1% and 5.9% of these state a posterior belief corresponding to the diagnosticity or the prior respectively. Our experiment therefore documents that, in balls-and-urns tasks with binary elicitation, subjects rarely communicate using explicit posterior beliefs.

As in our main experiment, we identify the main arguments and use GPT-4 to annotate them in individual explanations. Figure K14 plots the frequency of each argument in correct and incorrect explanations in its left panels. Base-rate neglect is by far the most common argument, appearing in 74.4% of incorrect and 49.6% of correct explanations. In these explanations, orators focus primarily on the ball composition of the bags while ignoring the card-deck prior. Signal neglect, i.e. reliance on the card-deck prior while ignoring the ball signal, is less common but still more frequent among incorrect explanations (11.6% versus 8.1%). This reasoning-based evidence confirms previous research on the topic (Bordalo et al., 2025). We call *Intuitive Confirming* the argument that points out when both prior and signal favor the same bag, without explicit calculations: it is naturally almost absent among incorrect explanations (1.2%) but common among correct ones (23.7%). Other intuitive but non-calculative arguments appear at moderate rates in both groups (5.8% and 9.6%). Explicit probabilistic calculations are rare overall. Correct mathematical reasoning appears in only 4.8% of correct explanations and never among incorrect ones, whereas wrong mathematical reasoning occurs in 4.7% and 4.3%, respectively. Despite our large data collection, the right panel shows we are not sufficiently powered to analyze the effects of individual arguments on imitation. Point estimates suggest correct mathematical reasoning has a strong positive effect on imitation, while base-rate neglect has a negative one.

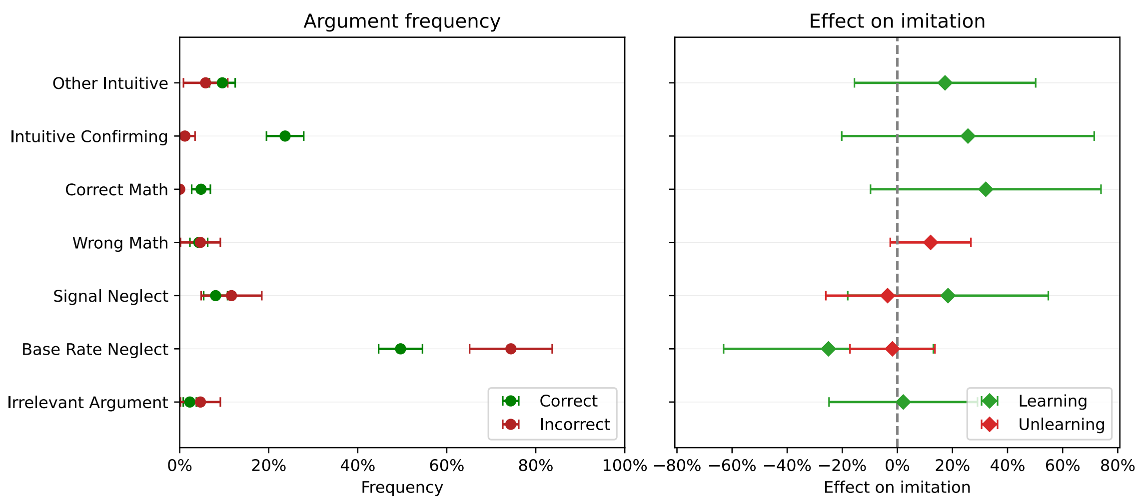


Figure K14: Argument frequencies and effects in binary Balls-and-urns experiment and effect on imitation. *Notes:* See notes for Figure 6. Sample is the Balls-and-urns receiver experiment (1,914 receivers and obs.). Whiskers show 95% CIs with clustering at the orator level.

Finally, we annotate the richness of explanations on a scale of 0-10. Correct explanations are 0.55 SD richer than incorrect explanations ($p < 0.01$): we thus document a significant richness gap in line with the main experiment. Table K7 then analyzes the effect of richness on imitation in learning and unlearning situations. Column (1) shows that there is a significant effect of *Explanation* on imitation only in learning situations, reflecting the main reduced-form finding. Column (3) shows that a 1 SD increase in richness increases imitation by 11.4 p.p. ($p < 0.01$). In turn, this means richness explains 24.9% of the learning asymmetry in the binary Balls-and-urns experiment. Other columns confirm this pattern weakens somewhat but remains qualitatively similar when controlling for the arguments used in each explanation, as well as receiver and orator characteristics. In conclusion, as in the main experiment, in the binary Balls-and-urns task respondents tend to imitate richer explanations more without imitating sparse explanations less, and correct explanations are richer on average, which drives their aggregate learning effect.

Table K7: Decomposition of differential learning effects in binary Balls-and-urns experiment

	Dependent variable: Imitation						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Explanation	-0.003	-0.243*	-0.340**	-0.308*	-0.534**	0.039	-0.449
	(0.036)	(0.139)	(0.172)	(0.181)	(0.264)	(0.039)	(0.274)
Learning	0.215***	0.259***	0.234***	0.173***	0.219***	0.231***	0.217***
	(0.065)	(0.083)	(0.067)	(0.064)	(0.074)	(0.068)	(0.075)
Explanation × Learning	0.249***	0.146	0.217**	0.305***	0.209**	0.187**	0.193*
	(0.087)	(0.112)	(0.095)	(0.084)	(0.104)	(0.090)	(0.106)
Richness						-0.028	0.016
						(0.035)	(0.039)
Explanation × Richness						0.114***	0.046
						(0.044)	(0.057)
Argument controls		✓			✓		✓
Orator controls			✓		✓		✓
Receiver controls				✓	✓		✓
Observations	532	532	532	532	532	532	532
R^2	0.224	0.277	0.268	0.316	0.384	0.247	0.389

Notes: See notes for Table 1. *Argument controls* denotes dummies for all arguments in Figure K14. As there was only one task in this experiment, orator and receiver controls do not include optimality on other tasks. Sample is the Balls-and-urns receiver experiment (1,914 receivers and obs.). Standard errors are clustered at the orator level.

L Continuous Balls-and-urns experiment

L.1 Logistics

The continuous balls-and-urns experiment resembles the binary one. We use priors of either 40% or 60%, and a diagnosticity held constant at 80%. However, the participant's task is to state the *likelihood* with which they think the blue-majority bag was selected. Beliefs are elicited using a text box. The continuous elicitation follows the behavioral literature on this task (Edwards, 1968; Bordalo et al., 2025).

Respondents in both studies received a base reward of \$2 for completing the study. Participants could win a bonus of \$10 following the same incentive structure as in the main experiment, defining an answer as correct if it is within ± 3 p.p. of the Bayesian posterior. Median completion times were 13 minutes in both experiments. Both experiments were run in April 2025 using a U.S. sample on Prolific. The Orator experiment was run for a total of 554 respondents, of whom 474 gave valid data. The Receiver experiment was run for 1,975 respondents, of whom 1,826 gave valid data. In line with our pre-registration, we drop respondents who self-reported using external tools. We further drop orators that a majority of receivers believed to have used an AI assistant: these cases represent 8.3% of valid recordings.

In light of prior research, we expected the share of respondents selecting a correct response to be low (Bordalo et al., 2025). This implies learning and unlearning situations should be very rare. As in the binary experiment, we therefore stratify the orator assignment based on the receiver's prior answer. If a receiver's prior answer is incorrect, we randomly assign them to a correct or incorrect orator with equal probability. If it is correct, we assign them an orator without stratification. In our raw sample, receiver and orator are both wrong in 40.6% and both right in 3.0% of observations, while learning and unlearning situations occur 43.0% and 13.5% of the time. We then de-stratify our sample using expected frequencies under random matching with an optimality rate of 16.7%, estimated by pooling orators and receiver priors. All summary statistics, regressions and figures are re-weighted to this population frequency.

L.2 Optimality

In line with the instructions and incentives given to survey participants, we define a correct answer as a belief within ± 3 p.p. of the Bayesian posterior. Figure L15 displays optimality rates for the balls-and-urns task. Prior optimality stands at 16.2% and 16.9% in *Choice Only* and *Explanation*, respectively ($p = 0.70$), which, as expected, is considerably lower than the optimality rates around 50% seen in the main experiment. As before, both treatments help listeners select correct answers and significantly increase their optimality rates by 3.9 p.p. ($p < 0.01$) and 3.3 p.p. ($p = 0.01$) respectively. However, the difference in posterior optimality between the two conditions is very small ($p = 0.97$).

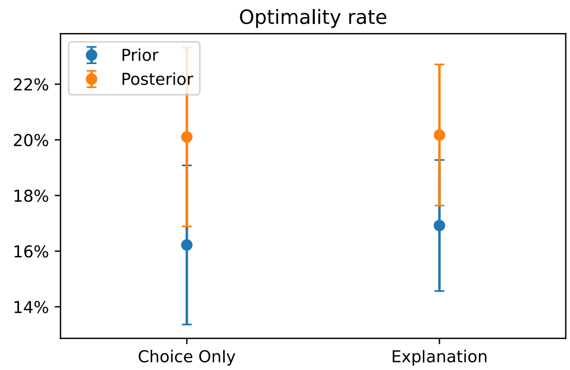


Figure L15: Effect of *Explanation* on optimality in Balls-and-urns experiment. *Notes:* Share of correct receivers before and after exposure to the orator's choice or explanation. An answer is defined as correct if it is within ±3 p.p. of the Bayesian posterior. Sample is the Balls-and-urns receiver experiment (1,826 receivers and obs.). Whiskers show 95% CIs with clustering at the orator level.

L.3 Imitation

In this abstract balls-and-urns setting, choices are continuous probabilities between 0% and 100%, so that imitation is harder to define and quantify. For completeness, we have therefore pre-registered four relevant measures of imitation:

1. *No imitation*: the receiver reports the same posterior belief as their prior belief

2. *Full imitation*: the receiver reports the same posterior belief as the orator

3. *Continuous imitation in [-1,2]*: to measure imitation continuously, we leverage a Grether (1980) decomposition to represent beliefs as log-odds in $\mathbb{R}$,[41] and then define continuous imitation as the interpolation weight placed on the orator's belief:

$$\text{Continuous imitation} = \frac{\text{logit}(\text{Receiver posterior belief}) - \text{logit}(\text{Receiver prior belief})}{\text{logit}(\text{Orator belief}) - \text{logit}(\text{Receiver prior belief})}$$

[41]Observations with beliefs of 0% or 100% are dropped from all analyses using continuous imitation.

No imitation yields a continuous imitation of 0, and full imitation yields 1. It can be negative if the receiver updates in the opposite direction, and larger than 1 if they go beyond the orator's belief. To limit the influence of extreme outliers due to beliefs close to 0% or 100%, we winsorize continuous imitation to $[-1, 2]$.

4. *Continuous imitation in [0,1]*: for completeness, we also report results for continuous imitation winsorized to $[0, 1]$.

Across all situations, the share of receivers who do not imitate at all is 50.8%, while 25.3% imitate fully. These two sets have a small overlap, as in 16.4% of cases the receiver has the exact same prior belief as the orator. These confirming signals are typically one of the salient modes of the problem, e.g., the prior or diagnosticity. On the other hand, 38.5% of receivers pick a posterior belief that is neither their own prior belief nor the orator's, with two thirds of these picking answers that are strictly between the two.

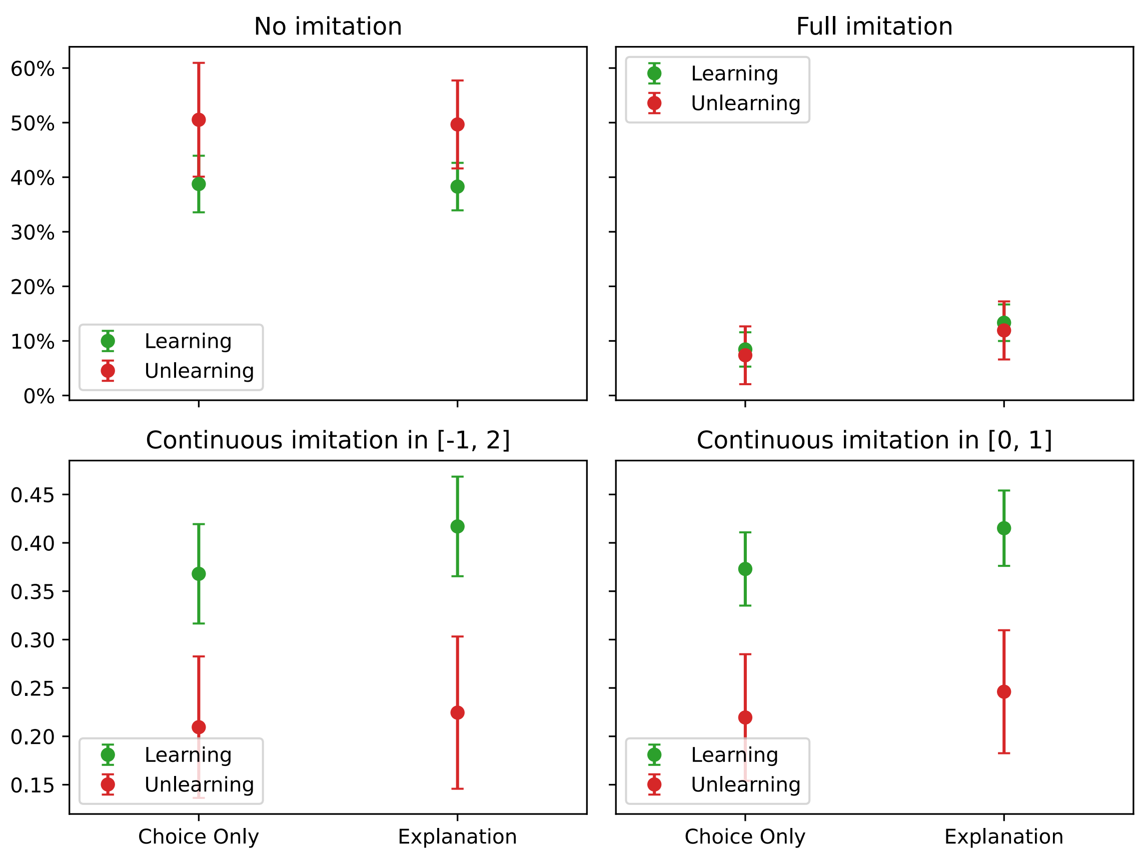


Figure L16: Effect of *Explanation* on imitation in Balls-and-urns experiment. *Notes:* See main text for definitions of imitation measures. Sample is the Balls-and-urns receiver experiment (1,826 receivers and obs.). Whiskers show 95% CIs with clustering at the orator level.

Figure L16 shows imitation in learning and unlearning situations. The share of receivers not imitating is close to 40% in learning situations for both treatments ($p = 0.89$) and closer to 50% for unlearning situations ($p = 0.90$). Full imitation rates are much lower, around 10%; they increase from 8.4% in *Choice Only* to 13.3% in *Explanation* in learning situations ($p = 0.03$), and from 7.4% to 11.9% in unlearning situations ($p = 0.25$). Continuous imitation winsorized to $[-1, 2]$ is about 40% in learning and 20% in unlearning situations, with insignificant differences between treatments ($p = 0.18$, $p = 0.78$). Findings are qualitatively similar for winsorization to $[0, 1]$ ($p = 0.10$, $p = 0.56$).

We further examine the full distribution of continuous imitation, as is done in Figure L17. The left panel shows the distribution pooling treatments, which displays strong modes at 0 and 1, corresponding to no and full imitation. There is a weaker mode around 0.5, suggesting that many receivers update somewhere roughly in the middle of their own prior belief and the orator's belief.[42] Imitation outside of these bounds is rare, as 12.7% of observations are outside of [0,1], and only 4.5% outside of [-1,2]. Splitting by treatments, the right panel reveals that the difference between *Choice Only* and *Explanation* appears small and unlikely to be statistically meaningful.

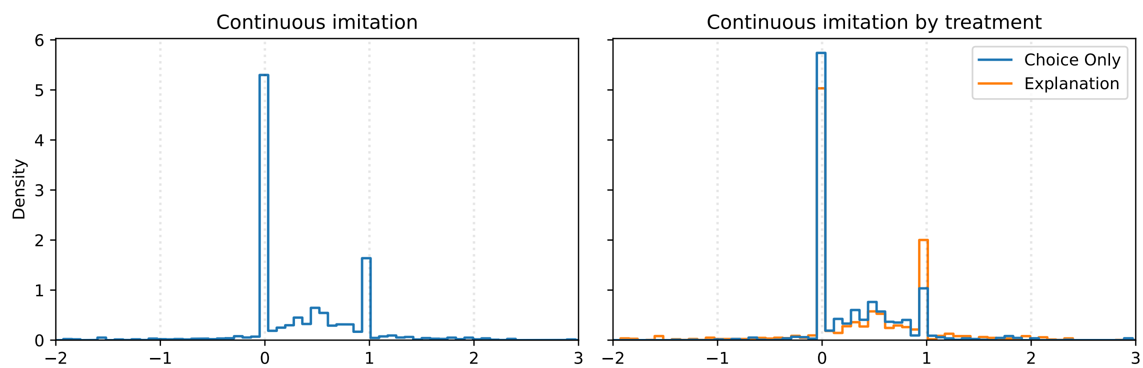


Figure L17: Distribution of continuous imitation in Balls-and-urns experiment. *Notes:* Left panel shows histogram of continuous imitation across treatments. Right panel shows histogram of continuous imitation in *Choice Only* and *Explanation*. The x-axis is cut to [-2,3], leaving 2.4% of observations outside of axis limits. Sample is the Balls-and-urns receiver experiment. Whiskers show 95% CIs with clustering at the orator level.

In conclusion, we only find very limited evidence of a treatment effect of *Explanation* relative to *Choice Only* in this continuous Balls-and-urns task. However, despite our oversampling strategy designed to maximize power, standard errors remain relatively

[42]Note that, with integer percentage beliefs, it is impossible to achieve an imitation of exactly 0.5 or a similar round value in log-odds space.

large, so that this is not a very precise null. The difficulty of the task, the complexity of social learning in this context, and the difficulty in defining optimality and imitation make aggregate improvement and learning hard to measure.

L.4 Imitation of richness

Annotating the richness of explanations for this experiment reveals a 0.75 SD difference ($p < 0.01$) between correct and incorrect explanations.

We examine whether richness predicts imitation. Table L8 replicates Column (7) of Table 1, which tests whether richness predicts imitation within learning and unlearning situations. We find that richness has a negative but statistically insignificant effect on no imitation ($p = 0.21$). However, it has a statistically significant effect on full imitation, at +3.6 p.p. ($p = 0.05$), and on continuous imitation winsorized to $[-1, 2]$ and $[0, 1]$, at +0.057 ($p = 0.07$) and +0.043 ($p = 0.06$) respectively. Hence, we do find evidence for our main hypothesized mechanism, the effect of richness on imitation, holds here. Crucially however, the magnitude of this effect is considerably smaller than for financial reasoning tasks.

Table L8: Imitation of richness in Balls-and-urns experiment

	No imitation (1)	Full imitation (2)	Continuous imitation (3)	Cont. in [0,1] (4)
Explanation	-0.012	0.050	0.021	0.031
	(0.066)	(0.039)	(0.054)	(0.046)
Learning	-0.110*	0.013	0.172***	0.155***
	(0.062)	(0.033)	(0.048)	(0.039)
Explanation × Learning	0.032	-0.024	-0.009	-0.017
	(0.078)	(0.047)	(0.070)	(0.054)
Richness	-0.011	-0.003	-0.019	-0.003
	(0.023)	(0.010)	(0.020)	(0.014)
Explanation × Richness	-0.035	0.036*	0.057*	0.043*
	(0.028)	(0.019)	(0.031)	(0.023)
Observations	1031	1031	1026	1026
R^2	0.020	0.014	0.036	0.053

Notes: Sample is the Balls-and-urns receiver experiment, restricted to learning and unlearning situations. *Explanation* is a dummy for the *Explanation* treatment, *Learning* is a dummy for learning situations. *Continuous imitation* is winsorized to [-1,2], *Cont. in [0,1]* to [0,1]. See main text for definition of imitation measures. Standard errors are clustered at the orator level.

Since the effect of explanations on optimality depends on both the richness gap and the responsiveness of receivers to richness, a useful back-of-the-envelope calculation is to combine the richness gap, the effect of richness on imitation, and the share of matches that generate learning or unlearning opportunities. In the continuous balls-and-urns task, the richness gap between correct and incorrect explanations is 0.75 SD, while a 1 SD increase in richness raises imitation by 3.6 p.p. (cf. column (2) of Table L8). This implies a predicted imitation effect of $0.75 \times 0.036 = 2.7$ p.p. Since the optimality rate stands at 16.7%, learning and unlearning situations occur in $2 \times 0.167 \times (1-0.167) =$ 27.8% of matches. Hence, the implied effect on optimality is just $0.75 \times 0.036 \times 0.278 =$ 0.8 p.p., well below the ±2.4 p.p. confidence interval in Figure L15.

In binary balls-and-urns, the richness gap is similar at 0.54 SD, but a 1 SD increase in richness raises imitation by a much larger 11.4 p.p. (cf. column (6) of Table K7). This implies an imitation effect of $0.54 \times 0.114 = 6.2$ p.p. With optimality at 82.2%, the expected frequency of learning and unlearning is $2 \times 0.822 \times (1-0.822) = 29.3\%$. Hence, the implied optimality effect is $0.54 \times 0.114 \times 0.293 = 1.8$ p.p., more than twice as large as in the continuous task (and matching our empirical effect of 1.8 p.p. exactly).[43]

Relative to the binary version, the continuous balls-and-urns treatment therefore has a much lower effect due to a much smaller imitation of richness and a lower frequency of (un)learning opportunities. On the common scale from 0 to 10, explanations in the binary balls-and-urns task have an average richness of 5.1, against 3.0 in the continuous balls-and-urns task, a significant and substantial difference ($p < 0.01$) even if comparisons across tasks is imperfect.

The null results in the balls-and-urns experiment are therefore largely a consequence of the environment rather than a failure of explanations. Correct explanations are richer in all experiments. When the task is made sufficiently tractable, richness becomes just as predictive of imitation as in financial reasoning tasks.

L.5 Imitation of salient modes

Previous research has shown that, when faced with abstract numerical problems like the balls-and-urns task, respondents rely on salient features to form an answer (Bordalo et al., 2025). In particular, experiments show that salience causes neglect of other relevant features, leading to biased answers that cluster around specific modes. For a balls-and-urns inference task similar to ours, Bordalo et al. (2025) find that respondents very often answer the prior or 100%-prior, the diagnosticity or 100%-diagnosticity, or fall back to

[43]For financial reasoning, the richness gap is 0.77 SD, while a 1 SD increase in richness raises imitation by 10.7 p.p. (cf. column (6) of Table 1), much higher than in binary balls-and-urns and close to the continuous version, implying an imitation effect of $0.77 \times 0.107 = 8.2$ p.p. With an optimality of 55.3%, the frequency of learning and unlearning is $2 \times 0.553 \times (1-0.553) = 49.4\%$. Hence, the implied effect on optimality is $0.77 \times 0.107 \times 0.494 = 4.1$ p.p. This is not far from the 3.2 p.p. effect in the data.

an ignorance default of 50%-50%. We obtain very similar modes in our data.

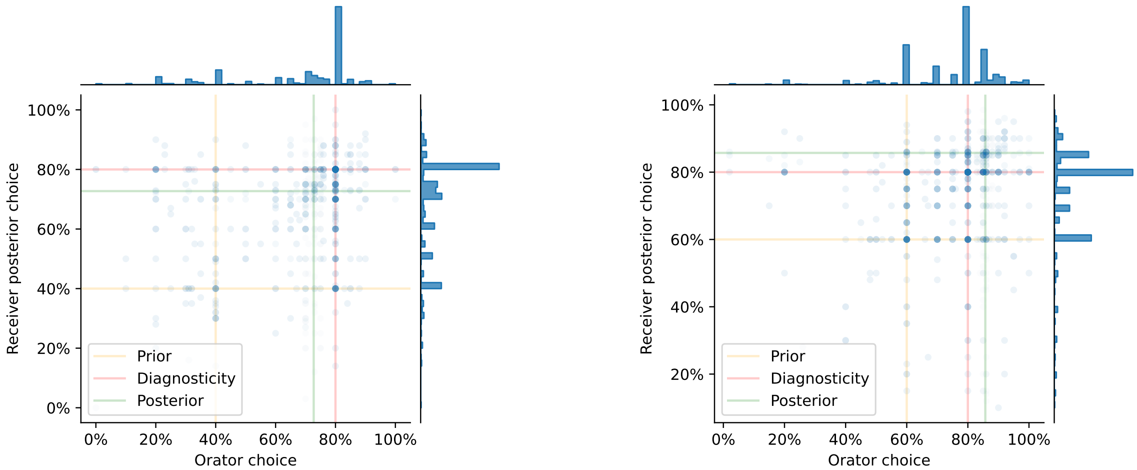
Our experiment allows us to study whether social learning about numerical problems also relies on salient features. We can examine whether other modes, besides the posterior, are more likely to be imitated. We confirm this approach is sensible by plotting orator beliefs and receiver posterior beliefs in Figure L18. To increase power, we have grouped symmetric configurations based on the color of the ball draw, so that there are only two groups: configurations with a prior of 40% on the color of the ball draw, and configurations with a prior of 60%. The left panel shows the first case, where the posterior of 72.73% is below the diagnosticity of 80%, and the right panel shows the second case, where the posterior of 85.71% is above. The marginal distributions of beliefs exhibit strong modes at the prior and the diagnosticity, and a weaker, more diffuse mode around the Bayesian posterior. The modes around 100%-prior, 100%-diagnosticity and 50%-50% are comparatively weaker.

The scatter plots, where the opacity of a dot reflects the number of overlapping observations, make clear that receiver posterior beliefs also strongly cluster on modes. Social learning is therefore largely based on switching from one mode to another, rather than smooth updating between likelihoods. For example, the dark blue dots along the "Diagnosticity" vertical line indicate a tendency by receivers to switch to this mode when faced with an orator that supports it. Salient features of a numerical problem therefore determine how subjects form responses to it, but also how they learn about it from others.

In line with our pre-registration, we identify orators' and receivers' reasoning errors directly from their estimated likelihood. Among orators, 12.5% respond with the prior, 36.9% answer the diagnosticity inverted according to the signal (e.g., if the diagnosticity is 80% and the ball draw is blue, the belief that the ball is red would be 20%), 16.7% give an answer within ± 3 p.p. of the Bayesian posterior, and 34.0% state some other belief. Orators responding 50%, the non-signal-flipped diagnosticity or the flipped prior are relatively rare in our data. Due to our oversampling scheme designed to increase the frequency of learning situations, we have limited power to study these rarer modes, and therefore group them into the "Other" category.

Receiver prior belief modes are very similar. Importantly, receiver posterior belief modes are also very similar. In particular, the share of answers not falling in one of the three modes is 35.3% and 35.1% respectively ($p = 0.89$), confirming that social learning heavily relies on modes.

Table L9 regresses imitation on the orator's choice mode, with "Prior" being the held out mode as it represents the simplest heuristic. It shows that "Diagnosticity · Signal" is



(a) Ball prior = 40%, Diagnosticity = 80%

(b) Ball prior = 60%, Diagnosticity = 80%

Figure L18: Distribution of orator and receiver beliefs in the continuous Balls-and-urns experiment. *Notes:* Symmetric configurations are grouped based on the color of the ball draw. Left panel shows configurations with a corresponding prior of 40%, right panel shows configurations with a corresponding prior of 60%. Histograms show marginal distribution, scatter plots show joint distribution, with opacity reflecting the number of observations. Sample is the continuous Balls-and-urns receiver experiment, pooling *Choice Only* and *Explanation* treatments.

Table L9: Imitation of modes in Balls-and-urns experiment

	No imitation	Full imitation	Continuous imitation	Cont. in [0,1]
	(1)	(2)	(3)	(4)
Intercept	0.454***	0.211***	0.410***	0.382***
	(0.038)	(0.036)	(0.041)	(0.032)
Diagnosticity · Signal	0.150***	0.260***	-0.048	-0.008
	(0.045)	(0.043)	(0.052)	(0.041)
Posterior ±3p.p.	-0.047	-0.061	-0.006	0.012
	(0.042)	(0.039)	(0.047)	(0.036)
Other	0.023	-0.125***	-0.136***	-0.096**
	(0.046)	(0.039)	(0.050)	(0.038)
Observations	1826	1826	1620	1620
R^2	0.023	0.153	0.011	0.014

Notes: Variables indicate orator belief mode: *Diagnosticity · Signal* is the diagnosticity inverted according to the signal draw, *Posterior ± 3p.p.* are beliefs within ± 3p.p. of the Bayesian posterior, *Other* are other answers. *Prior* is left-out. *Continuous imitation* is winsorized to [-1,2], *Cont. in [0,1]* to [0,1]. See main text for definition of imitation measures. Sample is the continuous Balls-and-urns receiver experiment, pooling *Choice Only* and *Explanation* treatments. Standard errors are clustered at the orator level.

significantly more likely to be fully imitated, with a 26.0 p.p. increase in full imitation, but also that it is more likely not to be imitated at all, with a somewhat smaller 15.0 p.p. increase in no imitation. Correspondingly, there is no effect on continuous imitation. Presumably, this is because this mode reflects an (erroneous) mathematical understanding of the problem, and therefore calls for complete agreement or (slightly more rarely) disagreement, but not smooth updating somewhere in-between, so that the full- and no-imitation effects cancel out.

We can further observe that "Posterior ± 3 p.p." has no significant effect on imitation, indicating it is not much more or much less convincing than "Prior", which is an interesting finding. Finally, "Other" has a strong negative effect on full and continuous imitation, indicating receivers are less prone to update fully or even partially based on orators that are not clearly on one of the dominant modes.

So far, we have pooled *Choice Only* and *Explanation* to maximize power. For completeness, Table L10 repeats the same analysis but distinguishes between *Choice Only* and *Explanation* by adding an interaction term for the latter. It finds little systematic evidence that *Explanation* changes the imitation of specific modes, in line with our previous findings for correct answers. However, it should be noted that our experiment is not very well-powered to study imitation of modes other than "Posterior ± 3 p.p." because of our stratification strategy, so that we can only detect relatively large effects.

Finally, we can examine the effect of the receiver's prior mode on imitation. Figure L19 plots the full and no-imitation rate by orator and receiver prior belief modes. Recall that two thirds of receivers either imitate fully or not at all, so that this Figure reflects transitions for an important fraction of the sample. Receivers who answered the prior and saw an orator who answered the prior maintained their answer in 69% of cases, a figure that goes up to 94% for diagnosticity signal, indicating that these are socially stable modes. On the other hand, only 33% of receivers who gave a correct answer fully imitate the orator, though this effect is partly mechanical because this mode covers a range of ± 3 p.p.. Indeed, a receiver who holds their belief while matched with an orator whose belief also falls within this range—but differs numerically—will not be classified as fully imitating, even if no belief updating occurs. The corresponding no-imitation rate is relatively high at 53%.

Transitions between modes are lower but still relatively high, typically around 10%-15%, and these smaller differences are unlikely to be significant. No-imitation rates for these tasks are correspondingly lower, around 50%.

On the other hand, orators giving an answer in the "Other" category see full-imitation rates closer to 5-10%, lower than the other modes, and no-imitation rates around 60%,

Table L10: Effect of *Explanation* on imitation of salient modes in Balls-and-urns experiment

	No imitation (1)	Full imitation (2)	Continuous imitation (3)	Cont. in [0,1] (4)
Intercept	0.579***	0.246***	0.336***	0.313***
	(0.069)	(0.057)	(0.071)	(0.059)
Explanation	-0.201**	-0.056	0.113	0.106
	(0.093)	(0.073)	(0.102)	(0.083)
Diagnosticity · Signal	0.046	0.206***	-0.002	0.014
	(0.079)	(0.069)	(0.083)	(0.070)
Explanation × Diagnosticity · Signal	0.166	0.088	-0.069	-0.029
	(0.105)	(0.089)	(0.118)	(0.097)
Posterior ±3p.p.	-0.149**	-0.102	0.024	0.052
	(0.074)	(0.063)	(0.076)	(0.063)
Explanation × Posterior ±3p.p.	0.163	0.066	-0.040	-0.057
	(0.100)	(0.080)	(0.112)	(0.089)
Other	-0.062	-0.219***	-0.075	-0.069
	(0.080)	(0.059)	(0.081)	(0.066)
Explanation × Other	0.137	0.151*	-0.092	-0.038
	(0.108)	(0.077)	(0.118)	(0.093)
Observations	1826	1826	1620	1620
R^2	0.030	0.157	0.013	0.022

Notes: See notes for Table L9. *Explanation* is a dummy for the *Explanation* treatment. Sample is the continuous Balls-and-urns receiver experiment. Standard errors are clustered at the orator level.

much higher than the others. Moreover, receivers whose prior belief was "Other" have no-imitation rates around 20%-30%, again considerably lower than for the other modes. They also appear more likely to imitate fully, in particular they fully imitate the "Diagnosticity · Signal" mode in 28% of cases. In conclusion, non-modal answers are more likely to be deviated from, and less likely to be imitated.

Summing up, the continuous Balls-and-urns task appears much harder, reflected in a much lower optimality rate, so that explanations do not increase aggregate optimality and show no robust effect across imitation measures. Analyzing the content of explanations and modal answers shows that reasoning errors are pervasive, making social learning effects much smaller and harder to detect. A comparison with the binary Balls-and-urns experiment, where the elicitation made the task easier and explanations did improve learning, suggests that explanations lose their power when tasks become too abstract or too complex. Bordalo et al. (2025) show that changing the framing of this task, for example by adopting the more naturalistic taxicab format (Kahneman and Tversky, 1972), changes the distribution of modal answers. It is therefore an open question, beyond the scope of this paper, whether framing complex tasks more naturally can lead to more social learning.

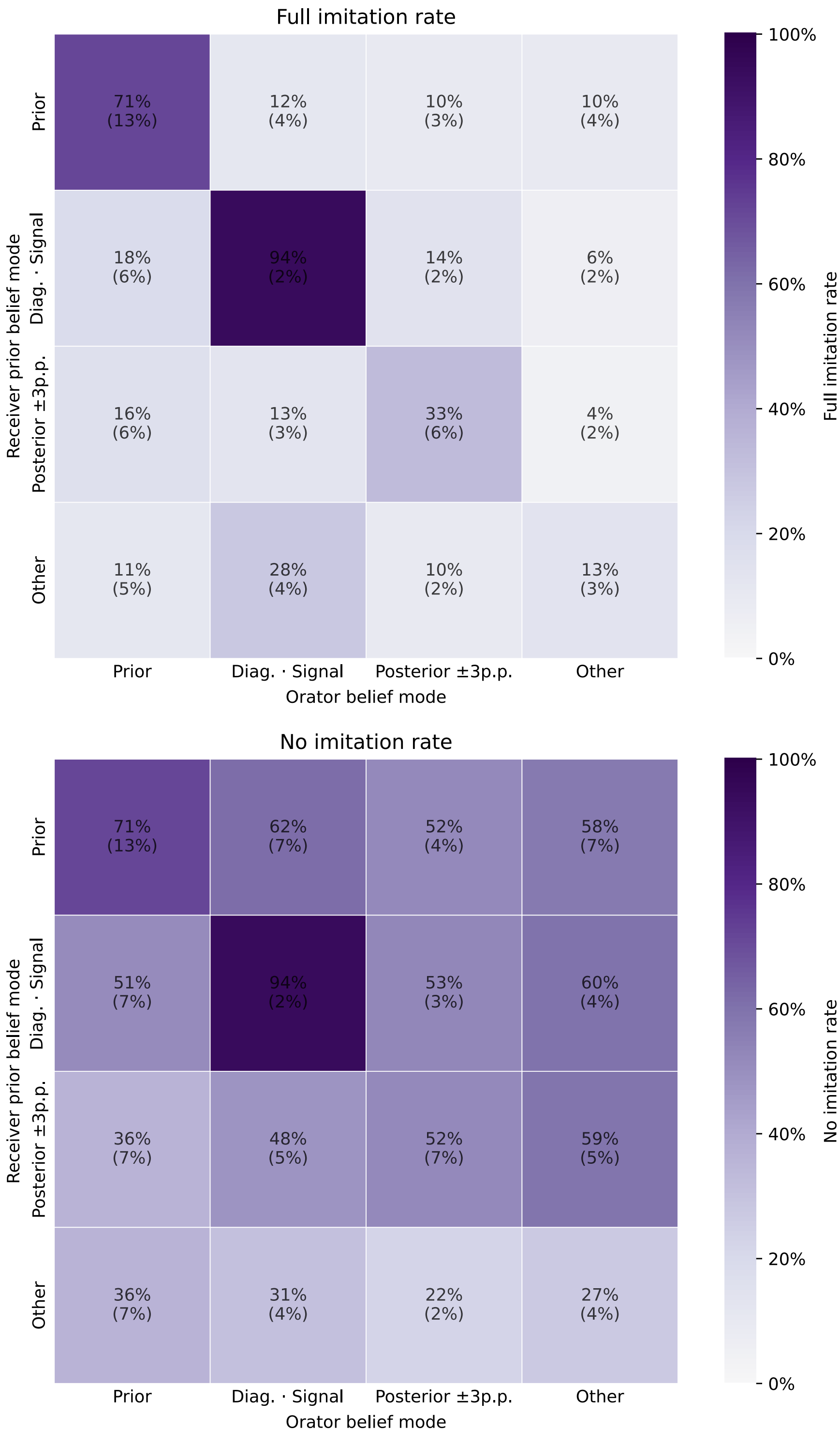


Figure L19: Imitation by receiver prior belief mode and orator belief mode in Balls-and-urns experiment. *Notes:* Top panel shows share of receivers stating the same belief as the orator, bottom panel shows share of receivers maintaining the same belief as their prior belief. Sample is the continuous Balls-and-urns receiver experiment.

M Survey screens

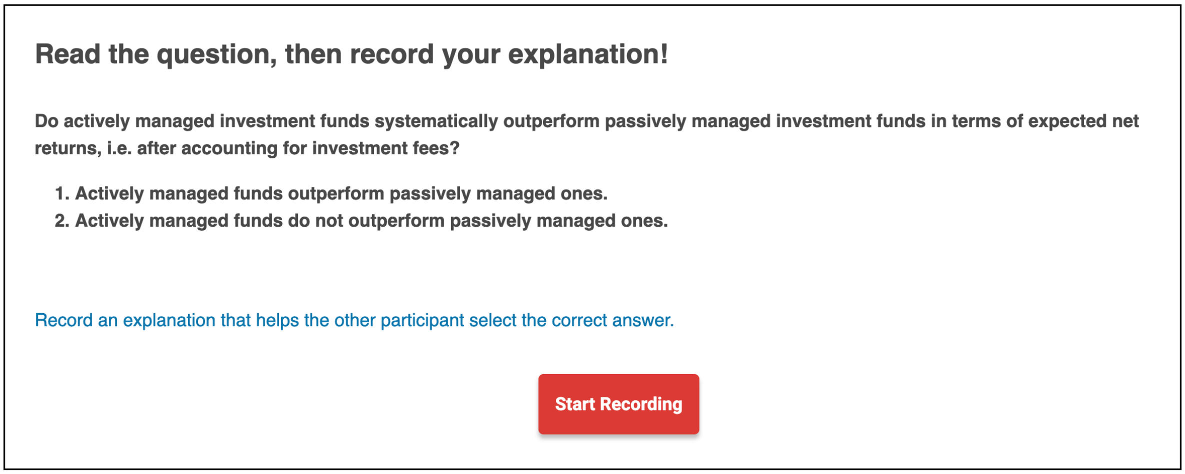


Figure M20: Recording screen from the Orator experiment.

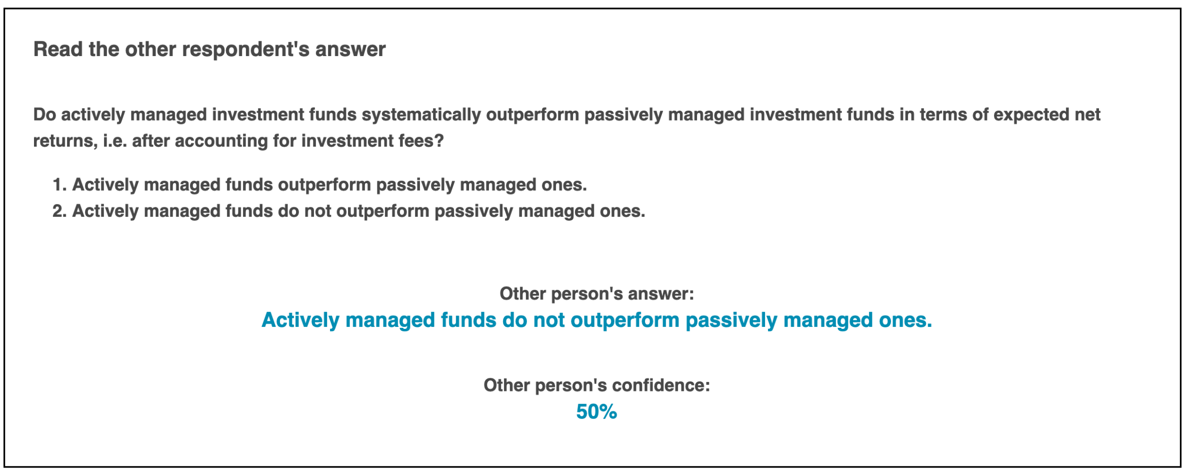


Figure M21: *Choice & Confidence* treatment screen from the Receiver experiment in Section 4.2.

Do actively managed investment funds systematically outperform passively managed investment funds in terms of expected net returns, i.e. after accounting for investment fees?

1. **Actively managed funds outperform passively managed ones.**
2. **Actively managed funds do not outperform passively managed ones.**

Other person's answer:

Actively managed funds outperform passively managed ones.

Other person's explanation:

All right. So I'm going to say that actively managed funds, um, actively managed funds do outperform, passively managed funds. And I'm going to say that which is an answer number one because I'm factoring in the level of risk management. So if there's risk management being actively applied to a, you know, to a, a fun then a lot of that risk that would just go on, you know, uncontrolled gets mitigated.

Figure M22: *Transcript* treatment screen from the Receiver experiment in Section 4.3.

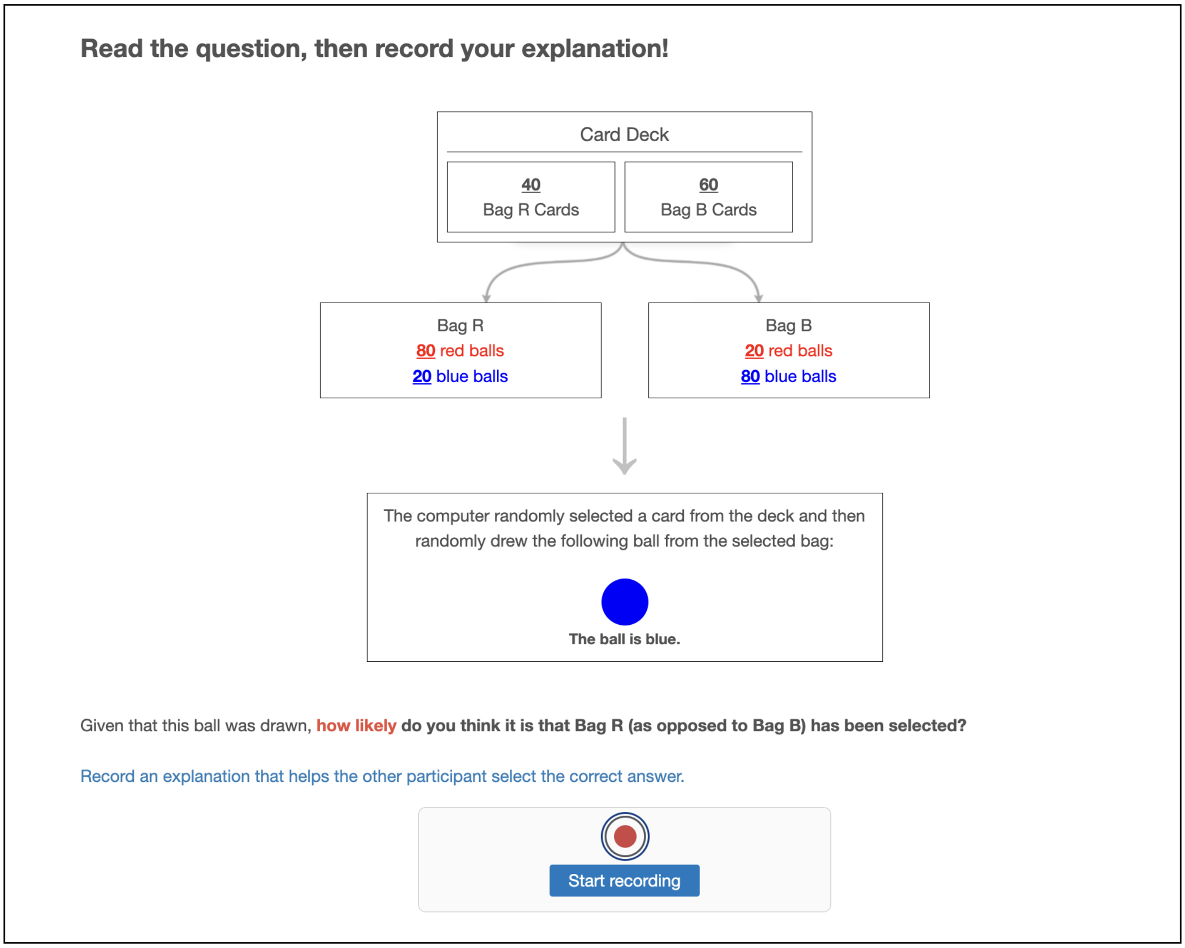


Figure M23: Recording screen from the continuous Balls-and-urns orator experiment.

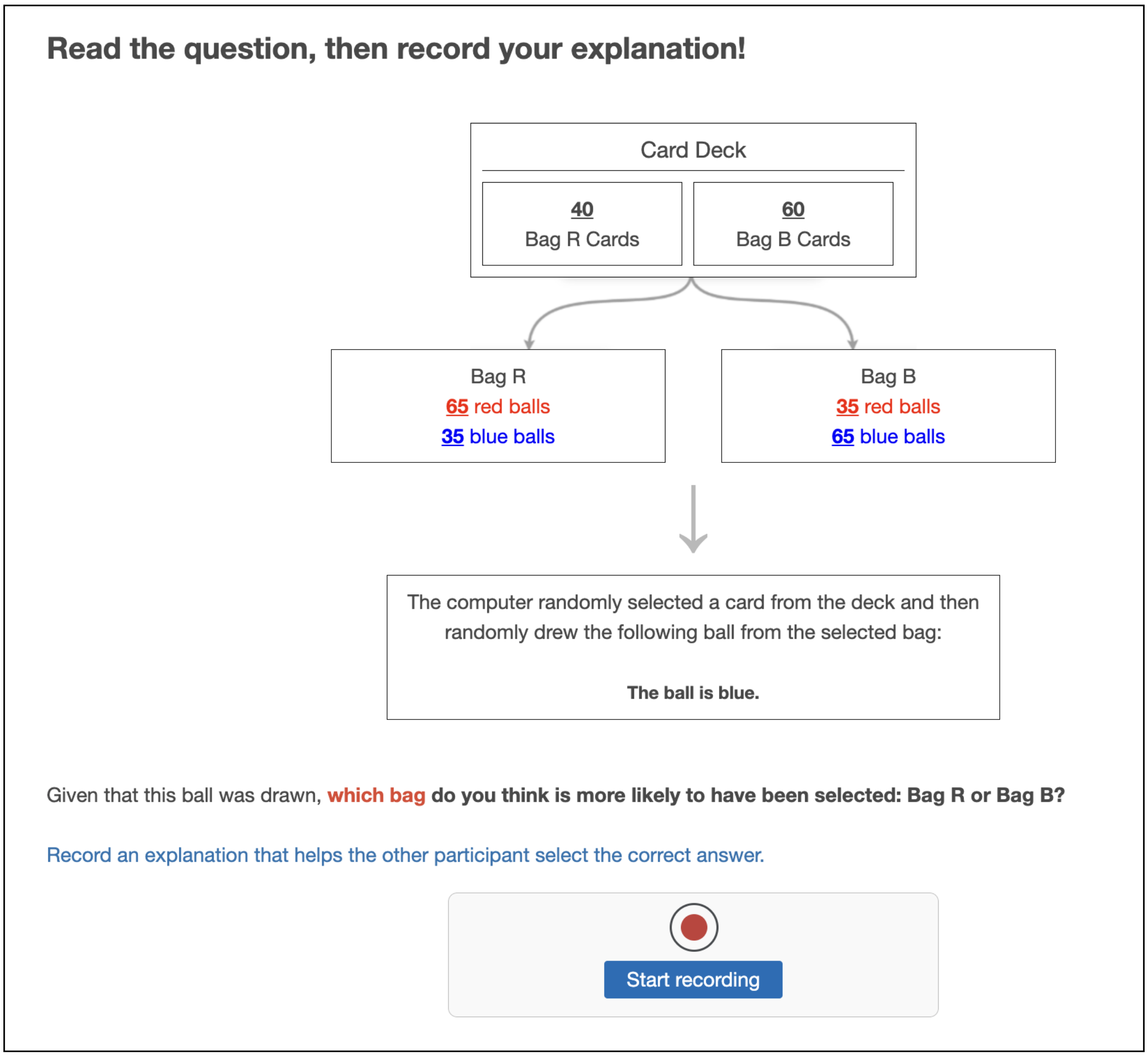


Figure M24: Recording screen from the binary Balls-and-urns receiver experiment.

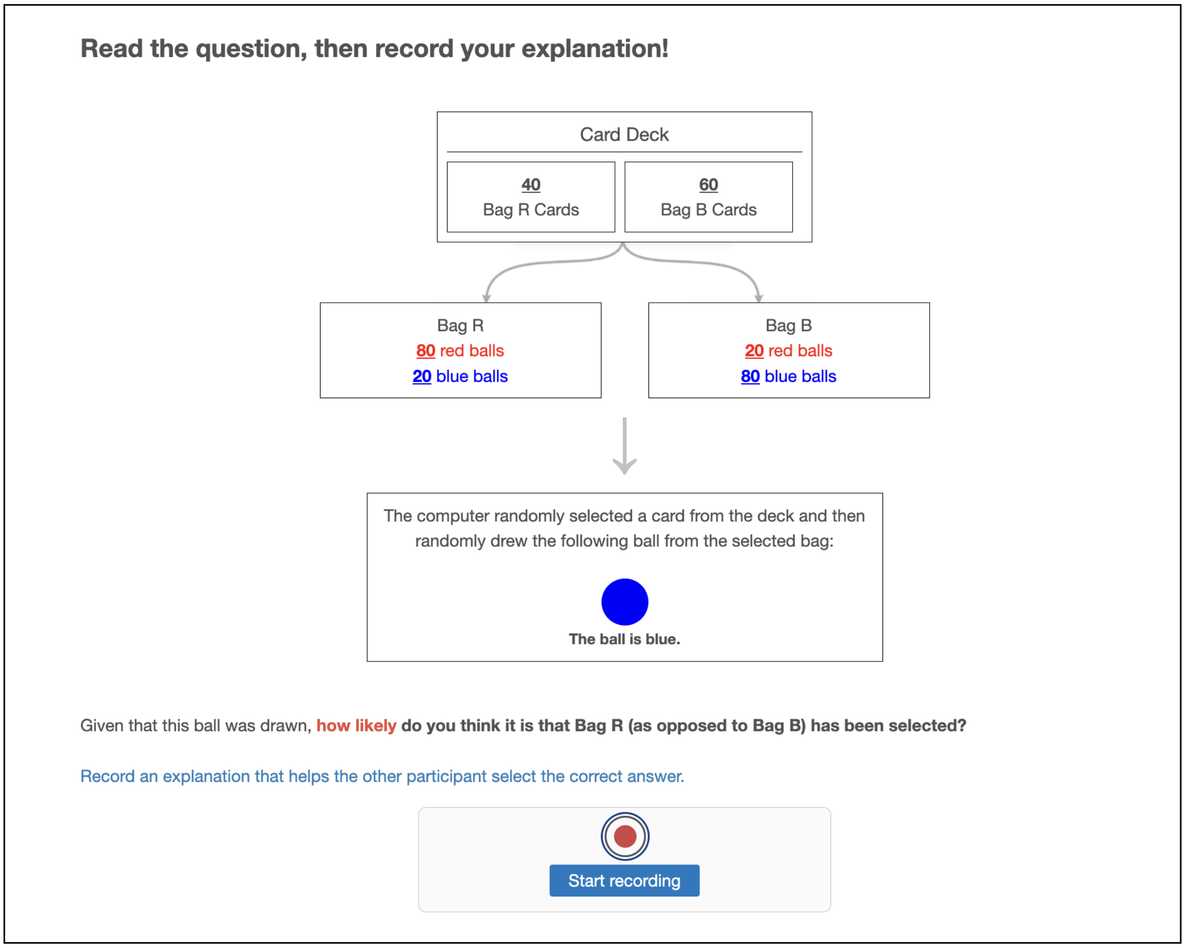


Figure M25: Recording screen from the continuous Balls-and-urns receiver experiment.

Question:

Do actively managed investment funds systematically outperform passively managed investment funds in terms of expected net returns, i.e. after accounting for investment fees?

1. **Actively managed funds outperform passively managed ones.**
2. **Actively managed funds do not outperform passively managed ones.**

Other person's answer:

Actively managed funds outperform passively managed ones.

Other person's explanation:

To answer this question, I chose that actively managed funds outperform passively managed ones, and my reasoning is that actively managed funds are, well, actively managed rather than passively managed. This means there's more hands-on involvement by the fund managers, as opposed to a passive approach. The idea is that this greater level of active management could lead to better performance, since someone is directly making decisions and adjustments, rather than just following a set index or strategy.

Which arguments did the person above use? Select every argument that appears in the explanation — this may be several, one, or none.

Actively managed funds can monitor and quickly adapt to market changes.

Human inability to predict market movements, performance pressure, errors or overconfidence limit the effectiveness of active management.

Expertise in active management can lead to better investment decisions.

Active managers get paid because clients expect them to bring higher results than passive funds.

References to historical data showing active management's performance.

References to historical data showing passive management's performance.

Passively managed funds maintain stability by not frequently changing investments, while actively managed funds are risky investments.

Passive management benefits from diversification across a broad market index.

Investment fees of actively managed funds are higher than for passive management. They reduce net returns and negate potential gains.

Passively managed funds tack market trends over the longer term, so that they are better at delivering long-term growth.

Passively managed funds can achieve long-term growth by following market trends. Passive management is efficient in tracking market performance with minimal intervention.

Any other substantive argument not part of the other categories.

Argument unrelated to the question; or no answer is implied by the argument.

Figure M26: Screen from the argument annotation experiment.